Bayesian Quantile Regression with Deep Echo State Network Features

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Abstract

Conditional quantiles are central to asymmetric decision losses, tail-risk assessment, and interval reporting, but Bayesian quantile regression for nonlinear dynamic series can be difficult when the temporal feature vector is high-dimensional. We develop the quantile deep echo state network (Q–DESN), a Bayesian quantile regression model conditional on a fixed deep echo state network feature map. Once the reservoir architecture, random seed, scaling, reducers, and washout convention are chosen, inference is restricted to the regression coefficients, likelihood, and shrinkage parameters. Single-level fits use asymmetric Laplace or quantile-fixed generalized asymmetric Laplace working likelihoods, with ridge or regularized-horseshoe priors and either Markov chain Monte Carlo or a model-specific variational Bayes approximation. For multi-quantile reporting, we distinguish independent level-wise regressions followed by a pre-specified monotone rearrangement from a joint quantile-vector regression that shares information through adjacent quantile-specific coefficient innovations. Synthetic studies, a one-origin retrospective GloFAS streamflow case, and a retrospective PriceFM comparison show when this fixed-feature Bayesian regression approach improves finite-grid quantile scores. The evidence supports limited claims: the reported bands are model-based quantile summaries, the finite-grid score denoted aCRPS is a trapezoidal finite-grid integral of check loss rather than full continuous-distribution CRPS, and calibration, crossing behavior, and reservoir sensitivity remain empirical diagnostics.

1 Introduction

Forecast evaluation often requires summaries beyond the conditional mean. Conditional means are natural under squared-error loss, but asymmetric decision losses, tail-risk assessment, and interval forecasts call for conditional quantiles. Under asymmetric piecewise-linear loss, conditional quantiles are optimal point forecasts, and pairs of quantiles define interval forecasts (Koenker and Bassett, 1978; Koenker, 2005; Gneiting, 2011). Let Y_{t+h} denote the future response at horizon h, and let $\mathcal{F}_t$ be the sigma-field generated by the information available at forecast origin t, including past responses and covariates known at the origin or specified through a scenario. With $F_{t,h}(u) = \Pr(Y_{t+h} \le u \mid \mathcal{F}_t)$, define the level-$p_0$ conditional quantile by the generalized inverse

$$Q_{t,h}(p_0) = F_{t,h}^{-1}(p_0) = \inf\{u : F_{t,h}(u) \ge p_0\}, \qquad p_0 \in (0,1).$$

A single quantile level is sufficient when the decision target is fixed. A grid of fitted levels can be post-processed into a monotone reported quantile curve, but that curve is a specified predictive summary rather than a full predictive distribution unless the analysis also specifies a joint predictive-distribution construction (Gneiting and Raftery, 2007; Gneiting and Katzfuss, 2014; Barlow and Brunk, 1972; Chernozhukov et al., 2010).

For nonlinear dynamic data, quantile forecast accuracy also depends on how temporal dependence is represented. Fixed reservoir features provide one computationally convenient representation: they encode recent history while leaving the recurrent weights outside the posterior inference problem. In the standard echo state network (ESN) construction used here, input and recurrent reservoir weights are generated once and then held fixed, so that only a regression layer, called the readout in the ESN literature, is estimated (Jaeger, 2001; Lukoševičius and Jaeger, 2009; Lukoševičius, 2012). A deep echo state network (DESN) extends this idea by stacking reservoir layers, following the deep ESN construction of Gallicchio et al. (2018). Conditional on the selected architecture, random seed, scaling constants, reducers, and washout convention, the resulting reservoir states define a deterministic nonlinear feature matrix. The goal in this article is conditional quantile estimation and forecasting for nonlinear dynamic series without placing posterior uncertainty on the recurrent network: the DESN supplies the fixed nonlinear feature map, and Bayesian inference is concentrated in the quantile regression model.

The quantile deep echo state network (Q–DESN) combines this fixed DESN design with Bayesian quantile regression. For a chosen level p_0, the linear predictor $\mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}$ is linked to the response through the quantile-fixed generalized asymmetric Laplace construction of Yan et al. (2025). To keep notation aligned with the exDQLM baseline and the exQDESN labels used in the validation tables, this article denotes that quantile-fixed extension of the asymmetric Laplace (AL) working likelihood by exAL. Thus an "ex" model prefix indicates the extended asymmetric Laplace working likelihood, whereas the unprefixed QDESN and DQLM labels indicate the AL working likelihood. Under this parameterization, μ_t is the model's p_0th conditional quantile, while the exAL asymmetry parameter allows mode, skewness, and tail behavior to vary relative to the asymmetric Laplace special case.

The AL or exAL likelihood is used as a working likelihood in the Bayesian quantile-regression sense. It defines a posterior distribution for quantile-specific coefficients without asserting that either likelihood is the data-generating error law (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). Posterior intervals for coefficient-derived quantiles and forecast bands generated from the fitted likelihood are therefore model-based summaries conditional on the fixed reservoir and the working likelihood. Predictive coverage is treated as an empirical diagnostic, assessed through simulation or validation data rather than assumed from the posterior alone.

High-dimensional regressions on DESN features require regularization. The default coefficient prior is ridge, which gives dense Gaussian regularization. As an adaptive alternative, the regularized horseshoe introduces global-local shrinkage and a slab component for large coefficients (Carvalho et al., 2010; Piironen and Vehtari, 2017). Under the product representation used here, the local, global, and slab scale updates retain closed-form inverse-gamma forms (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). Posterior computation includes an MCMC sampler for the Q–DESN hierarchy and a mean-field variational Bayes approximation for lower-cost screening, initialization, and selected application workflows. In exAL fits, the scale-asymmetry block is non-conjugate; the implementation uses the Laplace–Delta approximation for non-conjugate variational inference of Wang and Blei (2013) for that block. We use "VB–LD" only as implementation shorthand for this Wang–Blei-style approximation inside Q–DESN, not as a new variational-inference contribution.

The ESN literature calls the regression layer that maps reservoir features to the response scale a readout. Once the DESN features are fixed, inference in that layer is Bayesian quantile regression for the coefficient vector conditional on the fixed DESN design matrix. Throughout, RHS denotes the regularized horseshoe prior, MCMC denotes Markov chain Monte Carlo, VB denotes variational Bayes, and VB–LD denotes the model-specific variational Bayes approximation with Laplace–Delta moment updates.

This article makes three principal contributions. First, it introduces Q–DESN as Bayesian conditional-quantile regression on a fixed DESN feature map, with AL or exAL working likelihoods and ridge or regularized-horseshoe regularization. Second, it develops two distinct strategies for quantile grids: independent target-level regressions followed by a pre-specified monotone rearrangement, and a joint quantile-vector regression that shares information through regularized-horseshoe shrinkage on adjacent coefficient changes, borrowing from Quantile-Varying Parameter ideas. The joint prior encourages coherent quantile paths but does not impose hard noncrossing. Third, it provides MCMC and a model-specific Laplace–Delta variational approximation and evaluates their statistical and computational behavior in simulations plus two retrospective applications with explicit information-set limitations.

Construction	Inferential target	Fitting structure	Reported quantity
Single-level Q–DESN	$Q_{p_0}(Y_t \mid \mathcal{F}_t)$	AL or exAL working likelihood with ridge or RHS coefficient shrinkage; MCMC or VB	One fitted conditional quantile; working-likelihood simulation only under the pre-specified forecast protocol.
Independent-grid Q–DESN	$\{Q_{p_k}(Y_t \mid \mathcal{F}_t)\}_{k=1}^{K}$	Separate single-level fits at each target level, followed by a pre-specified monotone rearrangement	Quantile grid and interpolated reported quantile curve; no joint posterior over quantile-indexed coefficients.
Joint Q–DESN	Quantile vector over a fixed grid	Composite AL or exAL working likelihood with RHS shrinkage on baseline slopes and adjacent quantile-indexed slope increments; MCMC is the reported reference	Jointly estimated raw quantile path; exact monotonicity still requires a hard constraint or monotone rearrangement.

Table 1: Model map. AL versus exAL is a working-likelihood choice, ridge versus RHS is a coefficient-prior choice, independent versus joint fitting is a multi-quantile strategy, and MCMC versus variational Bayes is a computational choice.

What the empirical studies evaluate. The empirical studies address distinct questions. The single-quantile simulation compares QDESN and exQDESN regressions with DQLM and exDQLM baselines. The joint simulation evaluates multi-quantile coefficient sharing against independent level-wise regressions under known true conditional quantile paths. The GloFAS and PriceFM examples are retrospective applications of selected independent Q–DESN regressions; they should not be read as application-level evidence for the joint quantile-vector prior.

Related Work

Bayesian quantile regression provides the likelihood-based foundation for Q–DESN. Classical quantile regression targets conditional quantiles directly (Koenker and Bassett, 1978; Koenker, 2005). Bayesian implementations commonly use the asymmetric Laplace likelihood because of its connection to check-loss minimization and its mixture representation (Yu and Moyeed, 2001; Kozumi and Kobayashi, 2011). In this role, the asymmetric Laplace likelihood is a working likelihood: it defines a quantile-targeted posterior and supports conditional updates, while the coverage of model-based credible intervals may require separate assessment or adjustment under misspecification (Sriram et al., 2013; Yang et al., 2016; Ji et al., 2025). The quantile-fixed generalized asymmetric Laplace distribution of Yan et al. (2025) addresses a specific limitation of the asymmetric Laplace family by allowing the error distribution to vary after the target percentile has been fixed; this is the distribution denoted by exAL in the rest of the article.

Dynamic quantile models provide statistical baselines. Dynamic quantile linear models combine dynamic linear modeling with Bayesian quantile regression and provide posterior computation for time-varying quantile paths (Gonçalves et al., 2020). The exDQLM of Barata et al. (2022) adds asymmetric likelihood flexibility for dynamic quantile inference and is implemented, together with the corresponding AL/DQLM path, in the CRAN package `exdqlm` (Barata et al., 2026). Q–DESN changes the predictor representation: instead of specifying a low-dimensional linear dynamic state, it conditions on fixed nonlinear reservoir states and places a Bayesian quantile regression on the resulting design matrix. The DQLM and exDQLM fits therefore serve as structured linear dynamic quantile baselines under the same article-level forecast protocol, rather than as a complete decomposition of representation, likelihood, and prior effects.

Reservoir computing and ESNs provide the nonlinear dynamic features used by Q–DESN. ESNs were introduced by Jaeger (2001) and later reviewed within the broader reservoir-computing literature by Lukoševičius and Jaeger (2009); practical guidance is given by Lukoševičius (2012). Leaky-integrator ESNs make the leak rate an explicit time-scale parameter, while analyses of the echo state property caution that spectral-radius heuristics should be checked rather than treated as sufficient stability guarantees (Jaeger et al., 2007; Yildiz et al., 2012). Broader reservoir-computing surveys and time-series reviews discuss their role in forecasting and applied dynamic modeling (Zhang and Vargas, 2023; Yan et al., 2024; Cardoso et al., 2024; Sun et al., 2024). Deep ESNs, topology variants, and reservoir hyperparameter tuning extend the basic construction by stacking reservoirs, changing connectivity, or selecting reservoir parameters (Gallicchio et al., 2018; Kim and King, 2020; Maat et al., 2018; Bai et al., 2023; Viehweg et al., 2025). These sources motivate the fixed-feature viewpoint; Q–DESN places the Bayesian quantile-regression model conditional on those features.

A related reservoir-computing literature studies uncertainty quantification and probabilistic forecasting. Deep ESNs with uncertainty quantification, probabilistic reservoir methods for load forecasting, activation-level uncertainty propagation, and Bayesian or state-space ESN formulations show that reservoir features can be embedded in probabilistic forecasting models (McDermott and Wikle, 2019; Guerra et al., 2023; Gandhi et al., 2018; Singh and Raman, 2025). This literature sets the boundary of the present contribution: uncertainty quantification for ESNs is already active. The narrower gap addressed here is quantile-specific Bayesian coefficient inference under AL or exAL working likelihoods with high-dimensional regularization.

Quantile-oriented ESN methods provide the closest forecasting comparison. Lv et al. (2016) use quantile-regression ESN ensembles to construct prediction intervals for blast-furnace gas generation, and Bonas et al. (2024) use penalized quantile regression to calibrate DESN forecasts for quasi-periodic climate processes. Q–DESN is complementary to these approaches. It does not treat the reservoir training or penalized-quantile calibration step as the Bayesian object; instead, conditional on the generated reservoir, it specifies an exAL regression whose posterior includes coefficients, likelihood scale, asymmetry, and shrinkage parameters. The joint extension further couples quantile-specific regression coefficients during inference, so raw multi-quantile paths can be made more coherent before any monotone reporting rule is applied.

Shrinkage priors enter because fixed reservoir designs can be high-dimensional and correlated. Ridge shrinkage provides the default dense Gaussian regularization. The regularized horseshoe, building on the horseshoe prior and its slab-regularized extension, is used as an adaptive global-local alternative (Carvalho et al., 2010; Piironen and Vehtari, 2017). Under the product representation used here, scale updates remain closed form (Nishimura and Suchard, 2023; Makalic and Schmidt, 2016). These priors regularize the coefficients; they should not be interpreted as guaranteeing feature-level interpretability without additional diagnostics.

Finally, a grid of quantile levels requires a distinction between post-processing and joint modeling. Separate Q–DESN fits can cross, and they do not learn dependence across quantile levels. Isotonic regression can project the fitted grid onto the monotone cone (Barlow and Brunk, 1972), and rearrangement methods provide a general approach to noncrossing quantile and probability curves (Chernozhukov et al., 2010). These operations repair the reported quantile curve after fitting; they do not define a joint posterior over quantile-indexed coefficients. Constrained quantile-regression methods instead impose non-crossing restrictions during estimation (Wu and Liu, 2009; Bondell et al., 2010). Recent work also shows that non-crossing constraints can be viewed as fused shrinkage across quantile-specific coefficients (Szendrei et al., 2025). Bayesian joint quantile models share information across levels through process priors, score-likelihood constructions, or smoothness penalties (Yang and Tokdar, 2017; Wu and Narisetty, 2021; Wang and Cai, 2024). The Quantile-Varying Parameter (QVP) construction of Kohns and Szendrei (2025) is especially relevant here because it places a state-space shrinkage prior on adjacent quantile-indexed coefficient changes. The joint Q–DESN extension below adapts that idea to fixed-feature regressions and uses regularized-horseshoe shrinkage on adjacent slope innovations. The resulting prior is a soft non-crossing mechanism rather than a hard constraint. In the validation study below, it substantially reduces raw crossings relative to independent AL regressions before the monotone reporting rule is applied; exact monotonicity still requires an explicit ordered parameterization, projection, or rearrangement.

These distinctions determine the notation used below. We first separate the objects fixed by reservoir construction from the stochastic coefficient and likelihood quantities, so that posterior statements can be read as conditional on a recorded feature map.

The remainder of the paper is organized as follows. Section 2 introduces notation, deep echo state network (DESN) dynamics, and the exAL representation. Section 3 presents the Q–DESN model and prior specification, including independent and joint multi-quantile extensions. Section 4 describes posterior computation, while Sections 5–6 describe quantile reporting, monotone reporting rules, and reservoir diagnostics. Section 7 presents the simulation design, and Sections 8–9 give reproducible streamflow and electricity-price forecasting applications. Section 10 summarizes limitations and future work. The supplement gives full conditional derivations, variational and ELBO details, monotone reporting conventions, and additional validation tables.

2 Notation and Preliminaries

The notation separates deterministic reservoir quantities from stochastic coefficient and likelihood quantities. Table 2 summarizes the recurring objects. Matrices and vectors are bold, as in $\boldsymbol{W}$ and $\boldsymbol{x}$; scalar observations, latent states, and parameters use standard lowercase notation unless they are named constants. The symbols IG, GIG, and $\mathcal{N}^+$ denote inverse-gamma, generalized inverse Gaussian, and positive-truncated Normal distributions. For a square matrix $\boldsymbol{A}$, write $\rho(\boldsymbol{A}) = \max_i |\lambda_i(\boldsymbol{A})|$, where $\{\lambda_i(\boldsymbol{A})\}$ are the eigenvalues. Thus $\rho(\cdot)$ denotes spectral radius, while $\rho_d \in (0,1)$ denotes the target spectral radius for layer d. Indices use $d = 1, \ldots, D$ and $t = 1, \ldots, T$. When conditioning information is shown explicitly, $Q_p(y \mid \cdot)$ denotes the conditional p-quantile of the scalar response. The identity matrix, zero vector, and one vector of dimension m are denoted by $\boldsymbol{I}_m$, $\mathbf{0}_m$, and $\mathbf{1}_m$, respectively.

2.1 Deep Echo State Network (DESN)

The DESN is a deterministic feature map once the architecture, random seed, scaling constants, reducers, and washout convention are fixed. Conditional on these choices, posterior inference is restricted to the quantile regression model. The main dimensions are stated before the recursion because the hierarchy contains several layer-specific matrices.

Notation and dimensions. The input vector is $\boldsymbol{u}_t = (y_{t-1}, \ldots, y_{t-m}, \boldsymbol{z}_t^\top)^\top \in \mathbb{R}^{m+q}$, with m response lags and q exogenous variables. For layer d, $\boldsymbol{W}_d \in \mathbb{R}^{n_d \times n_d}$ is recurrent, $\boldsymbol{W}_1^{\text{in}} \in \mathbb{R}^{n_1 \times (m+q)}$ maps the original input to the first layer, and $\boldsymbol{W}_d^{\text{in}} \in \mathbb{R}^{n_d \times \tilde{n}_{d-1}}$ maps the reduced state from layer

Symbol	Description	Dimension and support
y_t	scalar response at time t	scalar
$p_0,\ p_k$	single-level and quantile-grid probabilities	$(0,1)$
$\boldsymbol{z}_t$	exogenous covariates	$q \times 1$
$\boldsymbol{u}_t$	response lags and exogenous covariates	$(m+q) \times 1$
$\boldsymbol{h}_{t,d}$	reservoir state at layer d	$n_d \times 1$
$\tilde{\boldsymbol{h}}_{t,d}$	reduced state at layer d	$\tilde{n}_d \times 1$
α_d, ρ_d	layer leak and target spectral radius	$[0,1]$ and $(0,1)$
$\tilde{\boldsymbol{x}}_t$	feature vector without intercept	$r \times 1$
$\boldsymbol{x}_t$	DESN design vector with intercept	$(r+1) \times 1$
$\boldsymbol{X}$	fixed DESN design matrix after washout	$T \times (r+1)$
$\boldsymbol{\beta}$	quantile-regression coefficients	$(r+1) \times 1$
κ_β^2	default ridge slope-prior variance	positive
σ, γ	exAL scale and asymmetry parameters	$\sigma > 0,\ \gamma \in (L, U)$
(λ_j, τ, ζ)	local, global, and slab shrinkage scales	positive

Table 2: Key notation used throughout. Reservoir quantities are drawn or constructed once, held fixed after washout, and collected in the design matrix $\boldsymbol{X}$.

$d-1$ to layer d for $d \geq 2$. The reducer $\boldsymbol{Q}_{d-1} \in \mathbb{R}^{\tilde{n}_{d-1} \times n_{d-1}}$ maps $\boldsymbol{h}_{t,d-1} \in \mathbb{R}^{n_{d-1}}$ to $\tilde{\boldsymbol{h}}_{t,d-1} \in \mathbb{R}^{\tilde{n}_{d-1}}$. The regression dimension is $r = n_D + \sum_{d=1}^{D-1} \tilde{n}_d + (m+q)$, and $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_r)^\top \in \mathbb{R}^{r+1}$.

Random sparse weights and spectral scaling. In contrast to fully trained recurrent networks, reservoir and input weights are drawn once and kept fixed throughout inference. Let $\circ$ denote the Hadamard product. Each recurrent matrix is $\boldsymbol{W}_d = \Phi_d^{(W)} \circ Z_d^{(W)}$, and each input matrix is generated analogously as $\boldsymbol{W}_d^{\text{in}} = \Phi_d^{(\text{in})} \circ Z_d^{(\text{in})}$. Mask entries are Bernoulli with probabilities $\pi_d^{(W)}$ and $\pi_d^{(\text{in})}$; base entries are independent draws from zero-mean finite-variance distributions p_W and p_{in}, such as $\mathcal{N}(0, \sigma^2)$ or $\text{Unif}[-a, a]$. Masks and base draws are mutually independent and independent across entries and layers. The recurrent matrices are then scaled to the target spectral radii,

$$\text{Spectral-radius scaling:} \qquad \bar{\boldsymbol{W}}_d \;=\; \frac{\rho_d}{\rho(\boldsymbol{W}_d)}\, \boldsymbol{W}_d, \qquad \rho_d \in (0,1).$$

If $\rho(\boldsymbol{W}_d) = 0$, the matrix is resampled. Input matrices $\boldsymbol{W}_d^{\text{in}}$ are not spectrally scaled.

DESN recursion and regression features. Figure 1 summarizes the three objects carried by the fixed feature construction: the lagged input, the layerwise reservoir update, and the fixed regression design.

With layer-specific leak rates $\alpha_d \in [0,1]$, elementwise nonlinearity $f(\cdot)$, and lower-layer transfor-

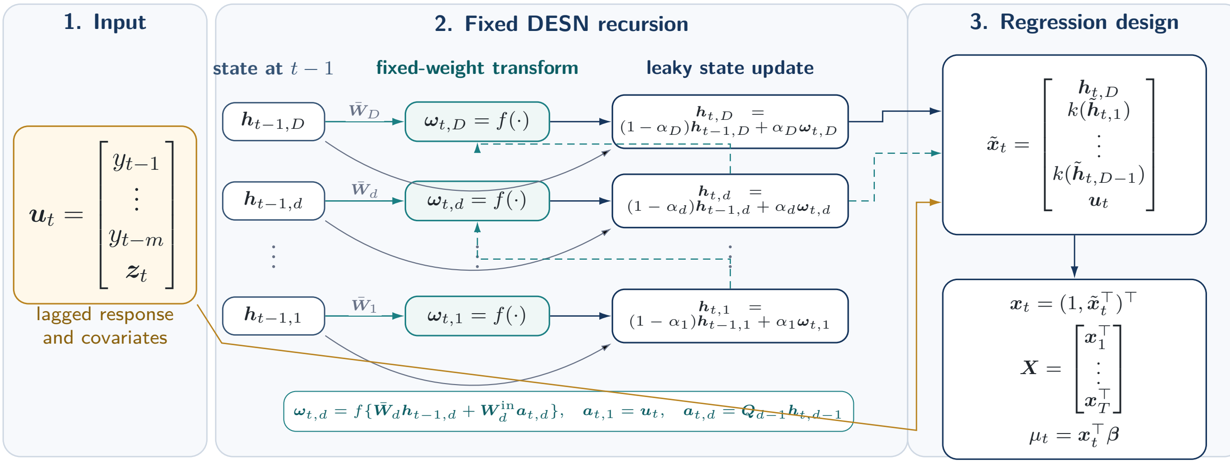

Figure 1: Fixed DESN feature construction. The input block collects response lags and exogenous covariates. The reservoir block shows that each layer update combines the previous state with a nonlinear innovation driven by fixed recurrent weights and, for upper layers, the current-time reduced state from the layer below. After washout, the top-layer state, transformed lower-layer reductions, and retained input block form the fixed regression design used by the Q–DESN working likelihood.

mation $k(\cdot)$, the fixed feature map is

$$
\begin{aligned}
\text{Layer 1 innovation}: &\quad \boldsymbol{\omega}_{t,1} = f\Big(\bar{\boldsymbol{W}}_1 \boldsymbol{h}_{t-1,1} + \boldsymbol{W}_1^{\text{in}} \boldsymbol{u}_t\Big), \\
\text{Layer 1 update}: &\quad \boldsymbol{h}_{t,1} = (1-\alpha_1)\boldsymbol{h}_{t-1,1} + \alpha_1 \boldsymbol{\omega}_{t,1}, \\
\text{Reduced bridge state}: &\quad \tilde{\boldsymbol{h}}_{t,d-1} = \boldsymbol{Q}_{d-1} \boldsymbol{h}_{t,d-1}, && d = 2,\ldots,D, \\
\text{Layer d innovation}: &\quad \boldsymbol{\omega}_{t,d} = f\Big(\bar{\boldsymbol{W}}_d \boldsymbol{h}_{t-1,d} + \boldsymbol{W}_d^{\text{in}} \tilde{\boldsymbol{h}}_{t,d-1}\Big), && d = 2,\ldots,D, \\
\text{Layer d update}: &\quad \boldsymbol{h}_{t,d} = (1-\alpha_d)\boldsymbol{h}_{t-1,d} + \alpha_d \boldsymbol{\omega}_{t,d}, && d = 2,\ldots,D, \\
\text{Feature stack}: &\quad \tilde{\boldsymbol{x}}_t = \big[\boldsymbol{h}_{t,D}^\top,\ k(\tilde{\boldsymbol{h}}_{t,1})^\top, \ldots, k(\tilde{\boldsymbol{h}}_{t,D-1})^\top,\ \boldsymbol{u}_t^\top\big]^\top, \\
\text{Intercept and linear predictor}: &\quad \boldsymbol{x}_t = (1, \tilde{\boldsymbol{x}}_t^\top)^\top, \qquad \mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}.
\end{aligned}
\tag{2.1}
$$

Here $k(\cdot)$ is an elementwise activation applied to reduced lower-layer states, commonly $k = \text{id}$ or $k = f$.

The input vector $\boldsymbol{u}_t$ bundles response lags and exogenous covariates; the intercept is appended only in $\boldsymbol{x}_t$. The leaky update combines persistence $(1-\alpha_d)\boldsymbol{h}_{t-1,d}$ with the nonlinear innovation $\boldsymbol{\omega}_{t,d}$. The leak rate α_d and target radius ρ_d control the empirical time scale and persistence of the deterministic state recursion; they are tuning choices rather than sufficient guarantees of the echo state property. Reducers $\boldsymbol{Q}_{d-1}$ may be fixed random projections, optionally orthonormalized, or precomputed linear reducers such as PCA from an initial preprocessing pass.

The feature stack concatenates the top-layer state, transformed reduced lower-layer states, and $\boldsymbol{u}_t$, yielding $\tilde{\boldsymbol{x}}_t \in \mathbb{R}^r$. Appending an intercept gives $\boldsymbol{x}_t \in \mathbb{R}^{r+1}$, and the coefficient vector maps this feature vector to the scalar location μ_t. States are initialized at $\boldsymbol{h}_{0,d} = \boldsymbol{0}_{n_d}$, or from $\mathcal{N}(\boldsymbol{0}_{n_d}, \sigma_{h,d}^2 \boldsymbol{I}_{n_d})$.

A teacher-forcing washout $t = 1, \dots, T_0$ is run before features enter the regression. Washout length and scaling choices can affect short-horizon behavior. All masks and base matrices are fixed for the entire analysis and are not resampled during posterior inference.

2.2 Extended Asymmetric Laplace Working Likelihood

The quantile-fixed generalized asymmetric Laplace distribution of Yan et al. (2025) extends the asymmetric Laplace (AL) working likelihood while keeping the target quantile fixed. We denote this likelihood by exAL, for extended asymmetric Laplace, to match the "ex" model prefix used in exQDESN and exDQLM. The notation is a naming convention for the Yan–Kottas quantile-fixed extension of AL, not a different distribution. The standard AL likelihood is recovered as a special case, but its skewness is fixed once p_0 is chosen. The exAL asymmetry parameter allows mode, skewness, and tail behavior to vary within a quantile-fixed likelihood. At fixed level p_0, μ is the p_0th quantile, $\sigma > 0$ is a scale parameter, and γ is an asymmetry parameter with bounded support (L, U). The endpoints L and U solve $g(\gamma) = 1 - p_0$ and $g(\gamma) = p_0$, respectively, where $g(\gamma) = 2\Phi(-|\gamma|)\exp(\gamma^2/2)$, and Φ is the standard Normal CDF.

Posterior computation uses the stochastic representation of the exAL distribution. If

$$y \sim \mathrm{exAL}_{p_0}(\mu, \sigma, \gamma),$$

then there exist independent $\epsilon \sim \mathcal{N}(0, 1)$, $z \sim \mathrm{Exp}(1)$, and $s \sim \mathcal{N}^+(0, 1)$ such that

$$y = \mu + \lambda(\gamma)\sigma s + A(\gamma)\sigma z + \{B(\gamma)\sigma^2 z\}^{1/2}\epsilon,$$

where $A(\gamma) = \{1 - 2p_\gamma\}/\{p_\gamma(1 - p_\gamma)\}$, $B(\gamma) = 2/\{p_\gamma(1 - p_\gamma)\}$, $C(\gamma) = \{\mathbf{1}\{\gamma > 0\} - p_\gamma\}^{-1}$, $p_\gamma = \mathbf{1}\{\gamma < 0\} + \{p_0 - \mathbf{1}\{\gamma < 0\}\}/g(\gamma)$, and $\lambda(\gamma) = C(\gamma)|\gamma|$. Setting $v = \sigma z$ gives $v \mid \sigma \sim \mathrm{Exp}(\mathrm{rate} = 1/\sigma)$ and the conditional Gaussian form used for Q–DESN posterior computation:

$$y \mid \mu, \sigma, \gamma, s, v \sim \mathcal{N}(\mu + \lambda(\gamma)\sigma s + A(\gamma)v,\ B(\gamma)\sigma v).$$

All quantities $g(\gamma)$, p_γ, $A(\gamma)$, $B(\gamma)$, $C(\gamma)$, and $\lambda(\gamma)$ depend on the target quantile level p_0; this dependence is suppressed for readability. Once (p_0, γ) is fixed, these are deterministic constants in the Gaussian augmentation.

3 Model Specification

Fix a quantile level $p_0 \in (0, 1)$. After reservoir construction and washout, $\boldsymbol{X} = (\boldsymbol{x}_1, \dots, \boldsymbol{x}_T)^\top$ is treated as fixed, and all posterior uncertainty is conditional on this design. The statistical model is a Bayesian quantile regression for $\boldsymbol{y}$ given $\boldsymbol{X}$. The observation model and priors define the posterior target; the Gaussian augmentation below is a computational representation of the asymmetric working likelihood. Dependence on p_0 is suppressed when no ambiguity arises.

3.1 Quantile Deep Echo State Network (Q–DESN)

The Q–DESN is obtained by inserting the fixed DESN feature map into the exAL working likelihood of Section 2.2. Let $\tilde{\boldsymbol{x}}_t$ be the feature vector in Eq. (2.1), and set $\boldsymbol{x}_t = (1, \tilde{\boldsymbol{x}}_t^\top)^\top$. Conditional on the fixed design matrix, the marginal working likelihood is

$$y_t \mid \boldsymbol{x}_t, \boldsymbol{\beta}, \sigma, \gamma \sim \text{exAL}_{p_0}(\boldsymbol{x}_t^\top \boldsymbol{\beta}, \sigma, \gamma), \qquad t = 1, \ldots, T.$$

Thus the dynamic linear predictor $\boldsymbol{x}_t^\top \boldsymbol{\beta}$ is the model's conditional p_0-quantile, while σ and γ control the scale and shape of the working error distribution. For posterior computation, the same exAL mixture representation is applied at each time point:

$$\begin{aligned}
\text{Quantile linear predictor}: &\quad \mu_t = \boldsymbol{x}_t^\top \boldsymbol{\beta}, \qquad \boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_r)^\top \in \mathbb{R}^{r+1}, \\
\text{Gaussian conditional}: &\quad y_t \mid \boldsymbol{\beta}, \sigma, \gamma, s_t, v_t, \boldsymbol{x}_t \sim \mathcal{N}\Big(\mu_t + \lambda(\gamma)\sigma s_t + A(\gamma) v_t,\ B(\gamma)\sigma v_t\Big), \\
\text{Positive-shift latent}: &\quad s_t \sim \mathcal{N}^+(0, 1), \\
\text{Scale-mixture latent}: &\quad v_t \mid \sigma \sim \text{Exp}(\text{rate} = 1/\sigma).
\end{aligned} \tag{3.1}$$

The display separates the statistical quantile model from the computational augmentation. The first line defines the quantile path; the second gives the conditional Gaussian observation equation; and the last two lines introduce auxiliary variables. The constants $A(\gamma)$, $B(\gamma)$, and $\lambda(\gamma)$ are inherited from Section 2.2. The DESN feature map is fixed before posterior inference, while $\boldsymbol{\beta}$, $\sigma > 0$, and $\gamma \in (L, U)$ are posterior unknowns. Integrating out (s_t, v_t) recovers the marginal exAL likelihood, so these variables do not alter the quantile target; they expose conditionally Gaussian updates. Complete-data joint distributions for the AL and exAL likelihoods with ridge and RHS priors are given in Section S4 of the supplement.

3.2 Single-Level Prior Specification

The intercept is modeled separately from the slope coefficients: $\beta_0 \sim \mathcal{N}(b_0, V_0)$, with $b_0 \in \mathbb{R}$ and $V_0 > 0$. This separation avoids forcing the baseline quantile level toward zero when the features are not centered around the target quantile.

Ridge prior. For the non-intercept coefficients $\boldsymbol{\beta}_{1:r} = (\beta_1, \ldots, \beta_r)^\top$, the default ridge prior is

$$\boldsymbol{\beta}_{1:r} \mid \kappa_\beta^2 \sim \mathcal{N}(\mathbf{0}_r,\ \kappa_\beta^2 \boldsymbol{I}_r), \tag{3.2}$$

where $\kappa_\beta^2 > 0$ is fixed or assigned a weakly informative scale prior. This Gaussian prior gives dense baseline regularization.

Adaptive shrinkage. The regularized horseshoe (RHS) prior is the adaptive shrinkage alternative. It starts from the ordinary horseshoe (Carvalho et al., 2010), adds slab regularization (Piironen and Vehtari, 2017), and uses the product representation of Nishimura and Suchard (2023) to retain closed-form scale updates. For non-intercept coefficients $j = 1, \dots, r$, the ordinary horseshoe is

$$\beta_j \mid \tau, \lambda_j \sim \mathcal{N}(0, \tau^2 \lambda_j^2), \qquad \lambda_j \sim C^+(0,1), \qquad \tau \sim C^+(0, \tau_0), \tag{3.3}$$

where τ is the global scale and λ_j is coefficient-specific. This prior concentrates mass near zero while allowing some coefficients to escape strong shrinkage, but it does not separately control the largest prior variances. The regularized version introduces a slab scale $\zeta > 0$ and sets

$$V_j(\lambda_j, \tau, \zeta) = \left(\zeta^{-2} + \tau^{-2}\lambda_j^{-2}\right)^{-1} = \frac{\zeta^2 \tau^2 \lambda_j^2}{\zeta^2 + \tau^2 \lambda_j^2}, \qquad j = 1, \dots, r, \tag{3.4}$$

with conditional prior

$$\beta_j \mid \lambda_j, \tau, \zeta \sim \mathcal{N}(0, V_j), \qquad j = 1, \dots, r. \tag{3.5}$$

This is the RHS coefficient prior used in the adaptive-shrinkage Q–DESN fits. Equation (3.4) gives the interpretation: if $\tau^2\lambda_j^2 \ll \zeta^2$, then $V_j \approx \tau^2 \lambda_j^2$, recovering ordinary horseshoe behavior; if $\tau^2 \lambda_j^2 \gg \zeta^2$, then $V_j \approx \zeta^2$, so the slab caps the effective prior variance.

The direct Piironen–Vehtari joint formulation gives the desired regularized horseshoe coefficient law. For Gibbs and CAVI updates, however, its global-local block includes an extra $\{1 + \tau^2\lambda_j^2/\zeta^2\}^{1/2}$ factor, so the ordinary horseshoe updates for (τ, λ_j) are not preserved. The Nishimura–Suchard joint construction preserves the same conditional coefficient law while restoring closed-form scale updates:

$$p(\beta_j, \lambda_j \mid \tau, \zeta) \propto \exp\left(-\frac{\beta_j^2}{2\tau^2\lambda_j^2}\right) \exp\left(-\frac{\beta_j^2}{2\zeta^2}\right) \lambda_j^{-1}(1 + \lambda_j^2)^{-1}, \tag{3.6}$$

where proportionality is with respect to (β_j, λ_j). The (τ, ζ)-dependent Gaussian normalizing factors are retained in the full-conditionals derivations so that the inverse-gamma shape parameters are recovered correctly. Thus the RHS option adds adaptive global-local shrinkage without losing inverse-gamma updates for the local and global scales.

Closed-form scale updates use the inverse-gamma representation of half-Cauchy priors (Makalic and Schmidt, 2016):

$$\lambda_j^2 \mid \nu_j \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\nu_j}\right), \qquad \nu_j \sim \mathrm{IG}\left(\frac{1}{2}, 1\right), \quad j = 1, \dots, r, \tag{3.7}$$

$$\tau^2 \mid \xi \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\xi}\right), \qquad \xi \sim \mathrm{IG}\left(\frac{1}{2}, \frac{1}{\tau_0^2}\right). \tag{3.8}$$

The slab scale is either fixed or assigned

$$\zeta^2 \sim \mathrm{IG}(a_\zeta, b_\zeta). \tag{3.9}$$

Under (3.6), this prior remains compatible with closed-form updating (Nishimura and Suchard, 2023).

The likelihood parameters are assigned $\sigma \sim \mathrm{IG}(a_\sigma, b_\sigma)$ and $\gamma \sim \pi_\gamma(\gamma)\,\mathbf{1}\{L < \gamma < U\}$, with (L, U) determined by the quantile-fixing construction as described in Section 2.2. For RHS fits, collect $\Theta = (\boldsymbol{\beta}, \sigma, \gamma, \boldsymbol{\lambda}^2, \tau^2, \boldsymbol{\nu}, \xi, \zeta^2)$, where $\boldsymbol{\lambda}^2 = (\lambda_1^2, \ldots, \lambda_r^2)^\top$ and $\boldsymbol{\nu} = (\nu_1, \ldots, \nu_r)^\top$. The entry ζ^2 is omitted when the slab scale is fixed. Ridge fits replace the global-local shrinkage block by κ_β^2 when that scale is assigned a prior.

3.3 Joint Quantile-Vector Regression with DESN Features

Joint quantile-regression target. The single-level model can be fit independently over a grid of quantile levels, but independent fitting does not couple regression coefficients across levels. For applications in which the coefficient path itself is part of the inferential target, let $0 < p_1 < \cdots < p_Q < 1$ be a fixed quantile grid and write $\boldsymbol{Z}$ for the $T \times r$ non-intercept fixed feature design. In a full Q–DESN analysis, $\boldsymbol{Z}$ is the non-intercept DESN design. The joint synthetic validation below uses the same regression model on a pre-specified design recorded by that validation study; it does not change the model definition. The joint quantile-vector regression separates intercepts from high-dimensional slopes:

$$Q_{p_q}(y_t \mid \mathcal{F}_t) = \alpha_q + \tilde{\boldsymbol{x}}_t^\top \boldsymbol{\beta}_q, \qquad q = 1, \ldots, Q, \quad t = 1, \ldots, T, \tag{3.10}$$

where $\alpha_q \in \mathbb{R}$ is the quantile-specific intercept and $\boldsymbol{\beta}_q \in \mathbb{R}^r$ is the slope vector at level p_q. The intercepts are not included in the high-dimensional RHS slope path by default; they are either weakly regularized independently or represented through ordered gaps when a stronger monotonicity convention is desired.

This construction generalizes the single-level RHS Q–DESN by replacing the single intercept-slope coefficient vector $(\beta_0, \boldsymbol{\beta}_{1:r})$ with $\{(\alpha_q, \boldsymbol{\beta}_q)\}_{q=1}^{Q}$. For $Q = 1$, the individual RHS Q–DESN is recovered by taking α_1 as the single-level intercept and identifying $\boldsymbol{\beta}_1$ with the single-level non-intercept slope vector, with no learned slope baseline. Equivalently, set $\boldsymbol{\beta}_{\text{base}} \equiv \mathbf{0}_r$ and place the RHS prior directly on $\Delta_1 = \boldsymbol{\beta}_1$.

Composite working likelihood. For each q, the working likelihood uses the exAL family at the corresponding target level,

$$y_t \mid \alpha_q, \boldsymbol{\beta}_q, \sigma_q, \gamma_q \sim \mathrm{exAL}_{p_q}\left(\alpha_q + \tilde{\boldsymbol{x}}_t^\top \boldsymbol{\beta}_q, \sigma_q, \gamma_q\right). \tag{3.11}$$

The Q likelihood contributions form a composite working likelihood over the same response series. All reported joint analyses use this untempered composite posterior target. It updates beliefs about the quantile-indexed coefficient path; it is not used as a normalized data-generating density for a future response.

Adjacent-innovation shrinkage. For a genuine multi-quantile grid, the joint prior adapts the Quantile-Varying Parameter (QVP) idea that adjacent quantile-indexed coefficient changes should be small unless the data support tail-specific effects (Kohns and Szendrei, 2025). Let $\boldsymbol{\beta}_{\text{base}} \in \mathbb{R}^r$ be a baseline slope vector and define adjacent innovations

$$\Delta_{1,j} = \beta_{1,j} - \beta_{\text{base},j}, \qquad \Delta_{q,j} = \beta_{q,j} - \beta_{q-1,j}, \quad q = 2, \ldots, Q.$$

The baseline slope receives an ordinary RHS prior, while the adjacent innovations receive RHS shrinkage with quantile-gap-specific global scales:

$$\beta_{\text{base},j} \mid \lambda_j^0, \tau^0, \zeta_0 \sim \mathcal{N}(0, V_j^0), \qquad V_j^0 = \frac{\zeta_0^2 (\tau^0)^2 (\lambda_j^0)^2}{\zeta_0^2 + (\tau^0)^2 (\lambda_j^0)^2}, \tag{3.12}$$

$$\Delta_{q,j} \mid \lambda_{q,j}^\Delta, \tau_q^\Delta, \zeta_\Delta \sim \mathcal{N}(0, V_{q,j}^\Delta), \qquad V_{q,j}^\Delta = \frac{\zeta_\Delta^2 (\tau_q^\Delta)^2 (\lambda_{q,j}^\Delta)^2}{\zeta_\Delta^2 + (\tau_q^\Delta)^2 (\lambda_{q,j}^\Delta)^2}. \tag{3.13}$$

This construction separates globally useful feature effects $\boldsymbol{\beta}_{\text{base}}$ from effects that vary across adjacent quantile levels. It is related to interquantile shrinkage and composite-quantile regularization in frequentist quantile regression (Jiang et al., 2014; Zou and Yuan, 2008), but the shrinkage is placed on the Bayesian quantile-coefficient path conditional on the fixed feature design.

The next display rewrites the Q augmented likelihood contributions as one stacked Gaussian block conditional on the exAL latent variables and the quantile-specific intercepts. Let $\boldsymbol{\beta}_{\text{st}} = (\boldsymbol{\beta}_1^\top, \ldots, \boldsymbol{\beta}_Q^\top)^\top$, $\boldsymbol{Z}_{\text{st}} = \boldsymbol{I}_Q \otimes \boldsymbol{Z}$, and $\boldsymbol{\alpha}_{\text{st}} = (\alpha_1 \mathbf{1}_T^\top, \ldots, \alpha_Q \mathbf{1}_T^\top)^\top$. Under the exAL augmentation, define

$$d_{q,t} = \lambda(\gamma_q)\sigma_q s_{q,t} + A(\gamma_q) v_{q,t}, \qquad \omega_{q,t} = B(\gamma_q)\sigma_q v_{q,t},$$

and stack these quantities into $\boldsymbol{d}$ and diagonal $\boldsymbol{\Omega} = \operatorname{diag}(\omega_{q,t})$. The conditional Gaussian likelihood is

$$\boldsymbol{y}_{\text{st}} \mid \boldsymbol{\beta}_{\text{st}}, \boldsymbol{\alpha}, \boldsymbol{d}, \boldsymbol{\Omega} \sim \mathcal{N}(\boldsymbol{\alpha}_{\text{st}} + \boldsymbol{Z}_{\text{st}}\boldsymbol{\beta}_{\text{st}} + \boldsymbol{d},\, \boldsymbol{\Omega}), \qquad \boldsymbol{y}_{\text{st}} = \mathbf{1}_Q \otimes \boldsymbol{y}.$$

Let $\boldsymbol{H}$ be the block first-difference matrix such that

$$\boldsymbol{H}\boldsymbol{\beta}_{\text{st}} = (\boldsymbol{\beta}_1^\top, (\boldsymbol{\beta}_2 - \boldsymbol{\beta}_1)^\top, \ldots, (\boldsymbol{\beta}_Q - \boldsymbol{\beta}_{Q-1})^\top)^\top.$$

With $\tilde{\boldsymbol{\beta}}_{\text{base}} = (\boldsymbol{\beta}_{\text{base}}^\top, \mathbf{0}_r^\top, \ldots, \mathbf{0}_r^\top)^\top$ and diagonal $\boldsymbol{D}_\Delta$ containing the RHS innovation variances $V_{q,j}^\Delta$, the conditional prior is proportional to

$$\exp\left[-\frac{1}{2}(\boldsymbol{H}\boldsymbol{\beta}_{\text{st}} - \tilde{\boldsymbol{\beta}}_{\text{base}})^\top \boldsymbol{D}_\Delta^{-1} (\boldsymbol{H}\boldsymbol{\beta}_{\text{st}} - \tilde{\boldsymbol{\beta}}_{\text{base}})\right].$$

Consequently, the Gaussian full conditional for the stacked slope path has precision

$$\boldsymbol{K}_\beta = \boldsymbol{Z}_{\text{st}}^\top \boldsymbol{\Omega}^{-1} \boldsymbol{Z}_{\text{st}} + \boldsymbol{H}^\top \boldsymbol{D}_\Delta^{-1} \boldsymbol{H}, \tag{3.14}$$

and mean

$$\boldsymbol{m}_\beta = \boldsymbol{K}_\beta^{-1} \left[\boldsymbol{Z}_{\text{st}}^\top \boldsymbol{\Omega}^{-1} (\boldsymbol{y}_{\text{st}} - \boldsymbol{\alpha}_{\text{st}} - \boldsymbol{d}) + \boldsymbol{H}^\top \boldsymbol{D}_\Delta^{-1} \tilde{\boldsymbol{\beta}}_{\text{base}}\right]. \tag{3.15}$$

The sparse, block-banded prior contribution in (3.14) is the central computational object for the joint extension.

Soft noncrossing. The non-crossing interpretation is local in adjacent quantile gaps. For neighboring levels,

$$Q_{p_q}(y_t \mid \mathcal{F}_t) - Q_{p_{q-1}}(y_t \mid \mathcal{F}_t) = (\alpha_q - \alpha_{q-1}) + \tilde{\boldsymbol{x}}_t^\top(\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}).$$

On a bounded feature domain, for example after an explicit scaling or clipping condition that puts each component of $\tilde{\boldsymbol{x}}_t$ in $[-1, 1]$, the inequality $\alpha_q - \alpha_{q-1} \geq \|\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}\|_1$ is sufficient for non-crossing. The adjacent-difference RHS prior does not impose this inequality. Instead, it shrinks $\boldsymbol{\beta}_q - \boldsymbol{\beta}_{q-1}$ toward zero, reducing the size of the feature-dependent term and concentrating the posterior near coherent quantile paths. Exact monotonicity still requires an explicit ordered-intercept parameterization, posterior rejection rule, isotonic projection, or monotone rearrangement. The synthetic joint validation below therefore records raw crossings separately from the pre-specified monotone rearrangement used for scored quantile grids.

4 Posterior Inference

4.1 Single-Level Augmented Posterior

Single-level target. For one target probability p_0, the Q–DESN posterior conditions on the fixed DESN design and updates only the regression coefficients, working-likelihood, and shrinkage parameters. Let $\boldsymbol{Z}$ denote the non-intercept design, α the intercept, and $\boldsymbol{\beta}_1$ the non-intercept slope vector. Let Θ_Δ collect the local, global, auxiliary, and slab scales for the RHS prior when the adaptive prior is used. For the exAL working likelihood, let $\{v_t, s_t : t = 1, \ldots, T\}$ be the scale-mixture and positive-shift latent variables; for AL, the s_t and γ blocks are omitted. With $\boldsymbol{d}_1$, $\boldsymbol{\Omega}_1$, and ω_t collecting the corresponding complete-data shifts and variances from the augmented likelihood, the single-level complete-data posterior can be written, up to constants not depending on unknowns, as

$$\begin{aligned} p(\alpha, \boldsymbol{\beta}_1, \sigma, \gamma, \{v_t, s_t\}_{t=1}^T, \Theta_\Delta \mid \boldsymbol{y}, \boldsymbol{Z}) &\propto \exp\left[-\frac{1}{2}(\boldsymbol{y} - \alpha \mathbf{1}_T - \boldsymbol{Z}\boldsymbol{\beta}_1 - \boldsymbol{d}_1)^\top \boldsymbol{\Omega}_1^{-1}(\boldsymbol{y} - \alpha \mathbf{1}_T - \boldsymbol{Z}\boldsymbol{\beta}_1 - \boldsymbol{d}_1)\right] \\ &\times \prod_{t=1}^T \left\{(\omega_t)^{-1/2} p(v_t \mid \sigma) p(s_t)\right\} p(\sigma)\pi_\gamma(\gamma)\mathbf{1}\{L < \gamma < U\} p(\alpha)\, p_\Delta^{(1)}(\boldsymbol{\beta}_1 \mid \boldsymbol{D}_\Delta) p(\Theta_\Delta). \end{aligned} \tag{4.1}$$

Here

$$p_\Delta^{(1)}(\boldsymbol{\beta}_1 \mid \boldsymbol{D}_\Delta) \propto \exp\left(-\frac{1}{2}\boldsymbol{\beta}_1^\top \boldsymbol{D}_\Delta^{-1} \boldsymbol{\beta}_1\right)$$

is the ordinary RHS slope prior, with $\Delta_1 = \boldsymbol{\beta}_1$. The ridge case replaces this factor by the Gaussian ridge prior. This display is the basic posterior target for the individual Q–DESN regression; the joint quantile-vector construction below reuses the same augmented Gaussian structure after stacking levels and replacing the direct slope prior by a regularized-horseshoe prior on quantile-indexed slope increments.

4.2 Joint Quantile-Vector Extension

Stacked joint target. For a grid $p_1 < \cdots < p_Q$, let Θ_Δ collect the RHS scales for adjacent slope innovations and let Θ_0 collect the baseline-slope RHS scales. Let Θ_Q collect $\boldsymbol{\alpha}$, $\boldsymbol{\beta}_{\mathrm{st}}$, the likelihood parameters, Θ_Δ, and, for $Q > 1$, $(\boldsymbol{\beta}_{\mathrm{base}}, \Theta_0)$. Let $\boldsymbol{a}_Q = \{v_{q,t}, s_{q,t} : q = 1, \ldots, Q;\ t = 1, \ldots, T\}$ collect the exAL auxiliary variables. With the stacked quantities $\boldsymbol{y}_{\mathrm{st}}$, $\boldsymbol{\alpha}_{\mathrm{st}}$, $\boldsymbol{Z}_{\mathrm{st}}$, $\boldsymbol{d}$, and $\boldsymbol{\Omega}$ defined in Section 3.3, the joint complete-data posterior is

$$\begin{aligned}
p(\Theta_Q, \boldsymbol{a}_Q \mid \boldsymbol{y}, \boldsymbol{Z}) &\propto \exp\left[-\frac{1}{2}(\boldsymbol{y}_{\mathrm{st}} - \boldsymbol{\alpha}_{\mathrm{st}} - \boldsymbol{Z}_{\mathrm{st}}\boldsymbol{\beta}_{\mathrm{st}} - \boldsymbol{d})^\top \boldsymbol{\Omega}^{-1}(\boldsymbol{y}_{\mathrm{st}} - \boldsymbol{\alpha}_{\mathrm{st}} - \boldsymbol{Z}_{\mathrm{st}}\boldsymbol{\beta}_{\mathrm{st}} - \boldsymbol{d})\right] \\
&\times \prod_{q=1}^{Q}\prod_{t=1}^{T}\left\{(\omega_{q,t})^{-1/2} p(v_{q,t} \mid \sigma_q) p(s_{q,t})\right\} \prod_{q=1}^{Q} p(\sigma_q)\pi_{\gamma_q}(\gamma_q)\mathbf{1}\{L_q < \gamma_q < U_q\} \\
&\times p(\boldsymbol{\alpha})\, p_\Delta^{(Q)}(\boldsymbol{\beta}_{\mathrm{st}} \mid \boldsymbol{\beta}_{\mathrm{base}}, \boldsymbol{D}_\Delta)\, p(\Theta_\Delta)\, \mathcal{B}_Q,
\end{aligned} \tag{4.2}$$

where

$$\mathcal{B}_Q = \begin{cases} p(\boldsymbol{\beta}_{\mathrm{base}} \mid \Theta_0) p(\Theta_0), & Q > 1, \\ 1, & Q = 1. \end{cases}$$

The Gaussian prior factor $p_\Delta^{(Q)}$ is induced by the first-difference matrix $\boldsymbol{H}$ and innovation covariance $\boldsymbol{D}_\Delta$ in (3.14)–(3.15). For $Q > 1$, it is the adjacent-innovation prior centered on the baseline slope, so the $\boldsymbol{\beta}_{\mathrm{base}}$ argument is active. The AL special case omits $s_{q,t}$, γ_q, and the positive-truncated Normal factors. Setting $Q = 1$, taking $\alpha_1 = \alpha$, $\boldsymbol{y}_{\mathrm{st}} = \boldsymbol{y}$, $\boldsymbol{Z}_{\mathrm{st}} = \boldsymbol{Z}$, $\boldsymbol{\alpha}_{\mathrm{st}} = \alpha\mathbf{1}_T$, $\boldsymbol{\beta}_{\mathrm{st}} = \boldsymbol{\beta}_1$, and dropping $\mathcal{B}_Q$ recovers the single-level posterior in (4.1). The exAL augmentation preserves the Gaussian slope-path update in (3.14)–(3.15), but the scale-asymmetry blocks (σ_q, γ_q) remain non-conjugate. We therefore use two computation routes for this fixed-design target: an MCMC sampler with separate exAL block updates and a mean-field variational Bayes approximation with Laplace–Delta updates for the non-conjugate blocks. In the algorithms below, GIG, positive-truncated Normal, Gaussian, and inverse-gamma entries denote closed-form full conditionals for MCMC and closed-form coordinate factors for VB. The same block structure reappears in the variational updates; only the final exAL scale-asymmetry block requires a non-conjugate treatment.

4.3 Markov Chain Monte Carlo

Sampler structure. The augmented likelihood and shrinkage hierarchy give a partially Gibbs sampler. Closed-form blocks are sampled from their named full conditionals; the complete-data targets, full conditionals, CAVI factors, and ELBO terms for AL and exAL likelihoods under ridge and RHS priors are given in Sections S5–S9 of the supplement. Under the Nishimura–Suchard hierarchy, the global-local scales have inverse-gamma updates. Each exAL block (σ_q, γ_q) remains non-conjugate and is updated with Metropolis–Hastings or slice moves on transformed coordinates. Algorithm 1 targets (4.2); with $Q = 1$, it targets the collapsed posterior in (4.1). Ridge and AL variants omit the corresponding shrinkage, $s_{q,t}$, and γ_q blocks. Diagnostics use trace plots, effective sample size summaries, crossing diagnostics for multi-quantile grids, and monitoring of the (σ_q, γ_q) updates, including acceptance rates when Metropolis–Hastings is used.

Algorithm 1 MCMC sampler for the Q–DESN posterior. For $Q = 1$, set $\boldsymbol{\beta}_{\text{base}} = \mathbf{0}_r$, place the RHS prior on $\Delta_1 = \boldsymbol{\beta}_1$, and omit baseline-slope blocks. AL and ridge variants omit the corresponding $s_{q,t}$, γ_q, and shrinkage blocks.

```
Require: Data {y_t, x̃_t}_{t=1}^T, quantile grid p_1 < ··· < p_Q, and priors
Require: MCMC iterations N and convergence-monitoring settings
Initialize α, β_st, likelihood parameters, latent variables, and RHS scales
if Q > 1 then
    Initialize β_base and its baseline RHS scales
end if
for i = 1 to N do
    Latent scales: for each q, t, draw v_{q,t} | · from the AL or exAL GIG full conditional
    exAL shifts: for exAL, draw s_{q,t} | · from its positive-truncated Normal full conditional; omit
        this step for AL
    Slope path: compute K_β and m_β from (3.14)–(3.15)
    Slope path: draw β_st | · ~ N(m_β, K_β^{-1})
    if Q > 1 then
        Baseline: draw β_base from its Gaussian conditional and draw its RHS scales from inverse-
            gamma full conditionals
    end if
    Intercepts: draw α from its Gaussian conditional, or update the ordered-gap block when
        exact intercept monotonicity is imposed
    RHS scales: draw innovation local, global, auxiliary, and slab scales from inverse-gamma full
        conditionals; for Q = 1, this is the ordinary single-level RHS block for β_1
    AL scale: for AL, draw each σ_q | · from its inverse-gamma full conditional and omit γ_q
    exAL scale–asymmetry: for exAL, update each (σ_q, γ_q) on transformed coordinates tar-
        geting its non-conjugate conditional kernel
    Monitor: store post-burn-in draws, crossing diagnostics, and convergence summaries, with
        thinning only if used
end for
```

4.4 Variational Approximation

VB–LD approximation. For lower-cost screening, warm starts, and companion validation panels, we also use a mean-field variational approximation to the same augmented target. The Gaussian, GIG, positive-truncated Normal, inverse-gamma, and RHS scale factors retain the closed-form coordinate structure implied by (4.2). The only extra non-conjugacy introduced by the exAL likelihood is the scale-asymmetry block (σ_q, γ_q). That block is approximated with the Laplace–Delta treatment for non-conjugate variational inference of Wang and Blei (2013), adapted to Q–DESN as detailed in the supplement. We therefore use the label VB–LD as a computational convention: it identifies the variational approximation used for screening and initialization, while the MCMC sampler remains the reference posterior computation when simulation tables report MCMC results.

Equations (4.2) and (4.1) define the reported posterior targets, and Algorithm 1 implements those targets. The supplement keeps the single-level full conditionals and the full quantile-vector posterior algebra explicit: Sections S5–S8 give the single-level posterior, MCMC, VB–LD, and ELBO updates, while Section S9 gives the joint quantile-vector posterior with regularized-horseshoe shrinkage over adjacent coefficient increments, stacked Gaussian blocks, and crossing diagnostics. For a single fitted quantile, Algorithm 1 uses $Q = 1$ and the collapsed RHS prior. For multiple fitted levels, the same sampler uses the baseline and adjacent-innovation blocks. Generic VB–LD factors, ELBO monitoring, the corresponding coordinate-update algorithm, and the monotone reporting rule are given in the supplement.

5 Multi-Step Quantile Forecasting and Scoring

Monotone rearrangement. Forecast propagation and multi-quantile reporting involve separate operations. The first operation propagates posterior or variational coefficient uncertainty through the fixed reservoir recursion at future horizons. The second, used only when several levels are reported together, applies a pre-specified monotone rearrangement or uses the joint quantile-vector regression introduced in Section 3.3. For a fitted grid $\widehat{\boldsymbol{q}}_t = (\widehat{q}_{1,t}, \ldots, \widehat{q}_{K,t})^\top$ at levels $p_1 < \cdots < p_K$, the default reporting rule is the weighted isotonic projection

$$\boldsymbol{q}_t^* = \arg\min_{q_1 \le \cdots \le q_K} \sum_{k=1}^{K} w_k (\widehat{q}_{k,t} - q_k)^2, \qquad w_k > 0, \tag{5.1}$$

with equal weights unless a validation protocol specifies another choice in advance. The ordered grid is then linearly interpolated over $[p_1, p_K]$. Outside the fitted range, the article reports grid-level quantile summaries rather than extrapolated tail claims unless an application explicitly states an endpoint convention.

5.1 Multi-Step Quantile Forecast Summaries

Pathwise quantile propagation. Reporting is conditional on the fitted quantile-regression coefficients, the fixed reservoir construction, and the working likelihood. At a forecast origin T, the historical reservoir state is obtained by teacher forcing: the recursion in Eq. (2.1) is run through the observed record with observed responses in the lag vector $\boldsymbol{u}_t$. After T, future responses are unobserved, so multi-step propagation is draw-specific. For a single-level fit at target p_0, write the draw as $\theta_{p_0}^{(m)} = (\boldsymbol{\beta}_{p_0}^{(m)}, \sigma_{p_0}^{(m)}, \gamma_{p_0}^{(m)})$. Conditional on a fixed future design row and this draw, the linear predictor

$$\mu_{p_0,T+h}^{(m)} = \boldsymbol{x}_{T+h}^{(m,p_0)\top} \boldsymbol{\beta}_{p_0}^{(m)}$$

is the draw-specific p_0-level linear predictor implied by the working model. If a response path is required for recursive lag updates, the fitted AL or exAL working likelihood can be used to generate

$$y_{T+h}^{(m,p_0)} \sim \mathrm{exAL}_{p_0}\left(\mu_{p_0,T+h}^{(m)}, \sigma_{p_0}^{(m)}, \gamma_{p_0}^{(m)}\right).$$

The resulting response draws are working-likelihood simulations. By contrast, the reported quantile summary for a fitted level is the pre-specified summary of $\mu_{p_0,T+h}^{(m)}$ or of the monotonized level-wise quantile grid. These summaries should not be conflated with a closed-form marginal quantile of $Y_{T+h} \mid \mathcal{F}_T$ unless a separate marginalization argument or simulation rule is supplied. Multi-step horizons add one further Monte Carlo step: future lagged responses inside the forecast window are generated or constructed pathwise, so future design rows are draw-specific.

Algorithm 2 Quantile forecast reporting for one fitted Q–DESN target level p_0.

Require: Target level p_0, forecast origin T, horizon H, and stored reservoir state

Require: Posterior or variational draws for the fitted p_0 model and origin-available or pre-specified scenario future covariates

```
Condition: teacher-force Eq. (2.1) through time T using observed responses and fixed reservoir weights
Notation: within the loop, write u_h^(m) = u_{T+h}^(m,p0) and x_h^(m) = x_{T+h}^(m,p0)
for each posterior or variational draw m do
  Initialize: set the reservoir state and response lags at the forecast origin
  for h = 1 to H do
    Design: form u_h^(m) from known covariates, observed pre-origin lags, and pathwise simulated forecast-window lags
    Reservoir: update Eq. (2.1) and form the draw-specific design vector x_h^(m)
    Linear predictor: compute μ_{p0,T+h}^(m) = x_h^(m)⊤ β_{p0}^(m)
    if a free-running response path is required then
      Emit: draw y_{T+h}^(m,p0) from the fitted AL or exAL_{p0} working likelihood conditional on μ_{p0,T+h}^(m)
    end if
    Report: retain μ_{p0,T+h}^(m) for p0-quantile reporting
  end for
end for
Aggregate: summarize quantile locations or emitted responses according to the pre-specified reporting rule; compute check loss, interval scores, aCRPS, and coverage diagnostics only for the corresponding reported summary
```

Scenario convention. Exogenous covariates must be known at the forecast origin, supplied as scenarios, or generated by an external covariate model. Additional details are given in the supplement. Thus, under independent single-level fits, "forecasting all levels" means applying

Algorithm 2 separately to each fitted p_k. A single p_0 fit should not be read as producing fitted target-level quantile summaries for all levels; off-target quantiles implied by its exAL working distribution are not the fitted quantile targets used in the monotone rearrangement.

There is a further distinction between level-wise marginal recursion and a synthesized reported path. If each fitted level is propagated separately, the resulting ordinates may condition on different simulated future response histories. When the ordinates are to be combined into one reported quantile curve, the recursion should instead use a common history: form all target ordinates from the same lag state, apply the monotone rearrangement, draw one $u \sim \text{Unif}(0,1)$, set $y = Q^*(u)$, and feed that same simulated response into every level-specific reservoir recursion at the next horizon.

For the joint quantile-vector regression, the reservoir recursion is unchanged. A posterior or variational draw contains the level-specific coefficients $\{\alpha_q, \boldsymbol{\beta}_q, \sigma_q, \gamma_q\}_{q=1}^{Q}$, so the same draw-specific future design vector returns a raw quantile vector over the grid $p_1, \ldots, p_Q$. That raw vector is the joint model output before any monotone reporting rule is applied. The composite quantile-vector likelihood should therefore be read as a device for estimating the conditional quantile path; it does not create a separate replicated future response for each quantile level. In the joint case, reporting over the grid uses the same conditioning, design, reservoir, and aggregation blocks as Algorithm 2, but replaces the single-level linear-predictor step by

$$\mu_{q,T+h}^{(m)} = \alpha_q^{(m)} + \tilde{\boldsymbol{x}}_{T+h}^{(m)\top} \boldsymbol{\beta}_q^{(m)}, \qquad q = 1, \ldots, Q,$$

and reports the resulting raw quantile vector, with optional monotone post-processing.

If recursive lag inputs require a scalar future response path, the joint model uses a quantile-curve sampling rule rather than draws from the composite working likelihood. Let $\boldsymbol{q}_{T+h}^{(m)} = (\mu_{1,T+h}^{(m)}, \ldots, \mu_{Q,T+h}^{(m)})^\top$. If exact monotone output is required, let $\mathcal{M}$ denote the pre-specified monotone rearrangement, such as isotonic projection or rearrangement (Barlow and Brunk, 1972; Chernozhukov et al., 2010), and set $\boldsymbol{q}_{T+h}^{*,(m)} = \mathcal{M}(\boldsymbol{q}_{T+h}^{(m)})$. The ordered grid defines a draw-specific reported quantile curve $Q_{T+h}^{*,(m)}(u)$ under the pre-specified interpolation and endpoint convention. A synthesized response value used for lag propagation is then

$$u_{T+h}^{(m,r)} \sim \text{Unif}(0,1), \qquad y_{T+h}^{(m,r)} = Q_{T+h}^{*,(m)}\left(u_{T+h}^{(m,r)}\right).$$

Thus the joint regression supplies the reported quantile curve used by the propagation rule; it should not be interpreted as drawing one future response per q, nor as mixing or normalizing the quantile-specific AL/exAL working likelihoods into a unique scalar predictive density.

6 Model Diagnostics and Reservoir Specification

Design record. Reported summaries and monotone reporting inherit the reservoir feature map selected before posterior fitting. Reservoir tuning, numerical diagnosis, and forecast evaluation therefore have distinct roles. Reservoir tuning chooses a reproducible feature-generation rule before the final analysis; numerical diagnostics screen candidate feature maps for alignment, stability, redundancy, and conditioning failures; forecast evaluation is defined by the simulation or application protocol. Because posterior inference is conditional on the fixed feature map, this design record is part of the empirical interpretation rather than a post hoc implementation detail. The simulation and streamflow sections specify the concrete fit-recovery and, where applicable, forecast-scoring criteria used in their empirical designs.

Let

$$\mathcal{R} = \{D, \{n_d\}, \{\tilde{n}_d\}, m, T_0, \{\alpha_d\}, \{\rho_d\}, \{\pi_d^{(W)}\}, \{\pi_d^{(\mathrm{in})}\}, f, k, \mathrm{seed}\}$$

denote the depth, layer sizes, reduction sizes, lag memory, washout, leak rates, spectral radii, sparsity probabilities, activations, and random seed used to construct Eq. (2.1). Once $\mathcal{R}$ and the preprocessing convention are fixed, posterior inference is conditional on the resulting design matrix $\boldsymbol{X}$. Specification selection chooses among candidate feature maps before the reported analysis; it does not average over reservoir architectures in the posterior target.

The ESN literature treats these choices as task-dependent design parameters. Reservoir size controls feature richness, leak rates and spectral radii jointly affect memory and stability, and input scaling determines how strongly the reservoir is driven into nonlinear regimes (Jaeger et al., 2007; Lukoševičius, 2012). For deep reservoirs, stacking layers can create hierarchical temporal representations, but additional depth must be justified by validation performance and stability diagnostics rather than by architecture alone (Gallicchio et al., 2018). The condition $\rho_d < 1$ is a useful conservative design choice, but it is not by itself a complete empirical check of the echo state property for a driven nonlinear reservoir (Yildiz et al., 2012). Several ESN modification papers identify failure modes that are visible before forecast scoring: output weights can reveal task-relevant reservoir units, state feedback changes the effective recurrent dynamics through an external feedback channel, redundant state trajectories can damage readout conditioning, and random reservoir realizations can induce substantial seed-to-seed variability (Fan et al., 2017; Ehlers et al., 2025; Saadat et al., 2025; Wu et al., 2018). In Q–DESN these results enter as diagnostics, sensitivity analyses, and possible ablations. They do not change the inferential target: after the feature map passes the validation protocol, the reservoir is fixed for posterior inference.

The protocol begins by fixing the empirical design before candidate reservoirs are compared: training and validation windows, held-out test windows, forecast origins, forecast horizons, quantile levels, input transformations, and scoring rules. Each candidate then passes a design check that verifies input alignment, feature dimensions, washout handling, finite state values, empirical spectral radii, state saturation, and readout conditioning. State-matrix diagnostics include near-zero-variance columns, pairwise state correlations, the singular-value or covariance spectrum, effective rank, and a condition number or eigenvalue-spread summary. These summaries distinguish a large reservoir that adds nonredundant dynamics from one that mainly adds collinear features. Candidates that fail the check are excluded before score ranking or sent to a remedial path, such as stronger shrinkage, dimension reduction, or state pruning.

Candidate ranking should combine design diagnostics with validation scores, but the relevant scores depend on the empirical setting. In simulations with a known oracle quantile path, fit-recovery metrics such as quantile-path RMSE, mean absolute error, bias, and correlation are available. In both simulations and applications, forecast scoring must use common forecast origins and horizons for all candidates. For a single fitted quantile level, check loss is the proper score aligned with the target functional. When several levels are fit and an estimated reported quantile curve is scored, interval score, empirical coverage, interval length, and aCRPS from the synthesized finite quantile grid may also be used, provided the fitted grid is dense enough for the summary being claimed. This aCRPS is twice the trapezoidal integral of check loss over the fitted levels; it is not full continuous-distribution CRPS unless an additional limiting or interpolation argument is established. These scores evaluate the reported single-level working-likelihood forecast or synthesized quantile grid, as applicable; they do not by themselves establish calibrated posterior credible bands under likelihood misspecification.

The coarse search focuses on depth, widths, lag memory, washout, leak rates, and spectral radii; the supplement records the fuller component-level reservoir checklist. Secondary choices such as recurrent sparsity, input sparsity, input scaling, and activations are broadened only when diagnostics indicate a specific failure mode. Top candidates are refit over multiple reservoir seeds and, when computationally feasible, compared under both VB and MCMC on a smaller diagnostic subset. The final reservoir specification is fixed before evaluation on held-out test origins. If reservoir ensembles are used, they aggregate only reservoirs that passed the same design checks and are reported as an ensemble sensitivity analysis rather than as posterior uncertainty conditional on a single fixed feature map.

Reproducible reporting includes the candidate grid, fixed defaults, state-quality summaries, rejected candidates and rejection reasons, validation origins, forecast horizons, scoring metrics, reservoir seeds, package and code versions, and the final selected $\mathcal{R}$. Claims about a selected architecture are conditional on this validation design. A single seed, a single forecast origin, or one favorable metric is not evidence of global optimality for the reservoir architecture. Reservoir-adaptation mechanisms such as state feedback or principal-neuron reinforcement are useful future extensions, but they change the feature-generation rule and require their own validation protocol before being combined with the Q–DESN regression.

7 Simulation Validation Study

The main simulation evidence is organized around two complementary questions. The first asks whether a single-quantile Q–DESN regression can recover and forecast a known dynamic quantile path under a 500-observation benchmark following the family-based comparison strategy of Yan et al. (2025). The target quantile level is fixed in advance, the data-generating mechanism is known, and competing models are compared under matched error families, fit-window sizes, and quantile levels. The second asks whether a joint multi-quantile regression can recover several known conditional quantile paths under posterior simulation and maintain coherent raw quantile grids over a fixed set of levels. Companion VB screening and forecast-comparison tables are kept in the supplement rather than used as the main article evidence.

In contrast to the linear-regression setting of Yan et al. (2025), the oracle quantile paths are dynamic, and Q–DESN represents those paths through high-dimensional deterministic reservoir features. The design therefore separates two questions: how accurately a fitted model recovers the known conditional quantile path, and how well it forecasts held-out dynamic blocks. We write p for the single target level in the first study and τ for levels in the multi-quantile grid.

The reported main-text tables are MCMC summaries from an archived validation record. That record fixes the simulated sources, rolling-origin metadata, fit and forecast windows, scoring rules, and table-generation scripts before any manuscript values are written. The supplement preserves VB companion panels and additional forecast-screening summaries. Larger-window MCMC studies and alternative simulation roots remain outside the present article tables. The joint multi-quantile validation below uses a separate fixed design and crossing-diagnostic protocol.

7.1 Single-Quantile Dynamic Data-Generating Processes

For each error family f and target quantile level $p \in \{0.05, 0.25, 0.50\}$, the synthetic series is generated as

$$y_{t,p,f} = \mu_{t,f} + \varepsilon_{t,p,f}, \qquad \varepsilon_{t,p,f} = e_{t,f} - H_f^{-1}(p),$$

where H_f is the distribution function of the raw error $e_{t,f}$. The shift by $H_f^{-1}(p)$ makes the p-quantile of $\varepsilon_{t,p,f}$ equal to zero. Hence

$$Q_p(y_{t,p,f} \mid \boldsymbol{\theta}_{t,f}) = \mu_{t,f}.$$

The oracle quantile path is generated by a six-dimensional dynamic linear state with local level, local slope, and two harmonic pairs:

$$\mu_{t,f} = \boldsymbol{F}^\top \boldsymbol{\theta}_{t,f}, \qquad \boldsymbol{F} = (1, 0, 1, 0, 1, 0)^\top,$$

with local linear trend and two sine–cosine harmonic pairs. The seasonal period is 90, the harmonics are 1 and 2, and the state innovation covariance $\boldsymbol{W}_\theta$ has diagonal square roots

$$\sqrt{\mathrm{diag}(\boldsymbol{W}_\theta)} = (0.005,\ 0.00002,\ 0.004,\ 0.004,\ 0.003,\ 0.003).$$

The initial covariance is $\boldsymbol{C}_0 = 0.01\boldsymbol{I}_6$, and each simulated trajectory is initialized deterministically at the family-specific state in Table 3. For a fixed family, the same latent path $\{\mu_{t,f}\}$ is used across the three values of p; only the centering constant $H_f^{-1}(p)$ changes.

7.2 Single-Quantile Fit and Forecast Protocol

Each source trajectory contains $T_{\mathrm{warm}} = 2000$ warmup values and $T_{\mathrm{main}} = 10000$ main source values. All indices below refer to the main source scale after warmup removal. The fitting origin is source

Family	Raw error distribution H_f	Initial level	Initial slope	Seasonal settings
Gaussian	$\mathcal{N}(0, 10^2)$	40	0.012	harmonics $(24, 0.35)$, $(8, -0.80)$
Laplace	Laplace with scale 10	35	0.011	harmonics $(28, -0.15)$, $(10, 0.75)$
Gaussian mixture	$0.1\mathcal{N}(0, 0.5^2) + 0.9\mathcal{N}(1, 15^2)$	45	0.014	harmonics $(32, 0.85)$, $(12, -1.25)$

Table 3: Single-quantile dynamic data-generating mechanisms. Raw errors are shifted by $H_f^{-1}(p)$ so that the p-quantile of the observation error is zero; the same latent dynamic quantile path is then shared across target levels within each family. Seasonal entries report amplitude and phase for the two harmonic pairs in the initial state.

index 9000, and the held-out block is 9001:10000. The two fit-window sizes are $T_{\text{fit}} = 500$ and $T_{\text{fit}} = 5000$, corresponding to training windows 8501:9000 and 4001:9000. Forecast evaluation uses a rolling-origin, no-refit state-update protocol over the held-out block. The primary protocol has maximum lead $H_{\max} = 30$ and origin stride 30. The reported tables use this pre-specified protocol and the validation record for source identity, model specification, and scoring metadata.

7.3 Single-Quantile Competing Methods

For each family and target quantile level in the single-quantile validation comparison, the reported Q–DESN rows vary the working likelihood and inference method while using regularized-horseshoe (RHS) coefficient shrinkage. Entries labeled Q–DESN AL–RHS use the AL working likelihood, whereas entries labeled Q–DESN exAL–RHS use the exAL working likelihood. Historical ridge fits are not included because they have not been reprocessed under the corrected train-only preprocessing rule. The reservoir specification, random seed, source identifiers, and software version for every displayed metric are retained in the reproducibility record. Dynamic quantile linear model (DQLM) and extended dynamic quantile linear model (exDQLM) baselines are fit to the same family, quantile level, fit-window, and forecast-origin protocol using `exdqlm` (version 1.0.0; Barata et al., 2026).

The primary comparison does not use repeated data-generating roots, a broad reservoir hyperparameter search, non-Bayesian ESN regression baselines, or post hoc multi-quantile synthesis. Those omissions are deliberate: the goal is to test whether Q–DESN regressions can be competitive with structured dynamic quantile linear models under linear dynamic mechanisms that favor the structured baselines.

7.4 Single-Quantile Criteria for Comparison

Let $\widehat{q}_t$ denote a posterior or variational summary of the fitted conditional p-quantile, such as the posterior median of the linear-predictor location. Because the oracle path μ_t is known, fit-window recovery can be evaluated using quantile mean absolute error, quantile RMSE, bias, correlation, and check loss at the target level p. Forecast accuracy is evaluated only on outcomes not used for fitting, under common forecast origins and leads for every model. The reported tables show fit recovery and forecast scoring; runtime, row-completion, and source-identity diagnostics are retained in the validation record.

For single-level quantile fits, the primary proper score is check loss at the fitted level. Quantile MAE and quantile RMSE compare the fitted or forecast quantile summary directly with the known oracle quantile path, while check loss scores realized observations relative to the reported quantile. These metrics therefore answer different questions and should not be collapsed into one rank. Runtime is retained as a computational criterion for comparing posterior simulation with VB in the reproducibility record and supplement, not as a statistical accuracy measure.

Because the simulation uses one reproducible dynamic root for each family and quantile level, the tables report deterministic criterion values for that source design rather than Monte Carlo means over repeated data sets. Any boldface or ranking in the displayed MCMC tables is interpreted within a fixed family, quantile level, fit-window, model family, and metric. The displayed Q–DESN values form a case-specific, metric-level summary: each entry is the best finite value observed for that model, family, quantile level, and metric. Consequently, entries within one model row may originate from different candidate specifications or replicate seeds. The contributing candidate specification, diagnostic status, and source identifier for every displayed entry are retained in the reproducibility record.

7.5 Single-Quantile Fit and Forecast Results

Evaluation protocol. The verified protocol uses the 500-observation training window 8501:9000, the held-out block 9001:10000, rolling-origin forecasts with leads 1–30 and origin stride 30, and the three target levels $p = 0.05, 0.25, 0.50$. Tables 4–6 report the selected MCMC family-specific metric summaries. The entries are generated from lead-level rolling-origin interface rows: fit RMSE measures recovery of the known oracle quantile path on the training window, forecast MAE measures oracle-quantile forecast error averaged across scored rolling-origin lead-target pairs, and forecast check loss scores realized observations at the fitted quantile level. Runtime and metric-level reproducibility information are retained separately. Boldface marks the lowest displayed MCMC value within each quantile level and metric, not a global ranking across different statistical criteria. Diagnostic status is disclosed rather than used as a hard metric-exclusion rule: daggers mark rows with a warning in the supporting diagnostic record, and double daggers mark rows with a failed diagnostic check.

All displayed Q–DESN and exQ–DESN values were recomputed after restricting feature scaling and every other learned preprocessing quantity to observations available through the fitting origin. The DQLM and exDQLM baselines use the same source trajectories, training window, forecast origins, leads, and scoring rules, but do not use the shared reservoir-scaling step.

For the exAL–RHS MCMC envelope, we additionally evaluated an exact v-augmented transition that collapses the scale from the shape update, redraws the scale exactly, and updates the shape on the logit transform of its native bounded support. Fifteen case-specific anchors were each evaluated with three independent chains, giving 60,000 retained draws per pooled anchor. A pooled criterion replaced its previously reported value only when it was strictly lower; this improved 22 of the 27 family–quantile–criterion entries and increased the number of exAL–RHS within-comparison minima from 10 to 16. The five non-improving entries retain their previous values and metric-level provenance. This confirmation changes the computational transition used for the new evidence, not the case-specific DESN or RHS specifications.

A subsequent paired confirmation revisited the Gaussian $p = 0.05$ and $p = 0.50$ exAL–RHS anchors using three independent full-budget chains per case. Under the pre-specified requirement that both the chain mean and median improve on the previously reported value, only the Gaussian $p = 0.05$ forecast MAE and check loss were replaced. Those two displayed criteria are arithmetic means over three chains with 20,000 retained draws each; fit RMSE and all $p = 0.50$ criteria retain their previous metric-level sources.

A forecast-first follow-up then revisited the Gaussian-mixture $p = 0.25$ exAL–RHS case under the exact scale-collapsed transition. Its three-chain means reduced forecast MAE from 3.396 to 1.819 and forecast check loss from 4.586 to 4.488; the fit criterion remained at its previous metric-level source. A subsequent adaptive forecast-gap campaign evaluated eight case-specific Q–DESN candidates with three full-budget chains per candidate. It replaced three additional forecast criteria: Gaussian-mixture $p = 0.50$ AL–RHS forecast MAE decreased from 2.368 to 0.907 and check loss from 5.585 to 5.441, while Gaussian $p = 0.05$ AL–RHS check loss decreased from 1.264 to 1.221. Every replacement is a strict three-chain mean improvement; no fit criterion changed. Diagnostic grades remain part of the provenance record but were not used to veto a lower finite forecast criterion.

Before regenerating the article tables, we also audited every Q–DESN forecast criterion against its archived lead-level rolling-origin paths. All 72 forecast metric roles, two for each of the 36 model–family–quantile–inference cells, were recoverable with no unresolved source, and 71 differed from the earlier scalar summary. The tables therefore use the values recomputed directly from those paths. This correction changes neither the selected case-specific models nor their fit-window recovery criteria, and the DQLM and exDQLM forecast summaries retain their verified rolling-origin metadata definition.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	3.036	3.829	1.103	3.811	2.209	3.328	2.590	**1.161**	**4.029**
exDQLM	**1.935**	16.23	4.313	2.344	6.634	4.132	2.808	1.637	4.109
Q–DESN AL–RHS†	2.779	8.410	1.221	2.176	**2.191**	**3.292**	1.523	2.104	4.073
Q–DESN exAL–RHS‡	2.149	**2.863**	**1.080**	**1.366**	2.415	3.317	**1.128**	1.971	4.059

Table 4: MCMC single-quantile fit-and-forecast comparison for the Gaussian simulation family. The Q–DESN entries form a fixed case-specific metric-level summary and can therefore draw different criteria from different calibrated fits. When repeated-chain results are available, the table uses the pre-specified repeated-chain aggregate identified in the metric-level record. Q–DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger marks a warning in the supporting diagnostic record; a double dagger marks a failed diagnostic check.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	**3.663**	9.791	1.968	2.850	3.520	4.548	1.774	1.337	5.082
exDQLM	5.677	22.44	5.154	1.710	6.811	5.106	1.774	2.013	5.206
Q–DESN AL–RHS†	5.420	3.893	1.873	2.256	1.434	4.379	1.353	0.7553	**5.008**
Q–DESN exAL–RHS†	6.409	**2.362**	**1.855**	**1.383**	**0.6696**	**4.343**	**1.075**	**0.6772**	5.008

Table 5: MCMC single-quantile fit-and-forecast comparison for the Laplace simulation family. The Q–DESN entries form a fixed case-specific metric-level summary and can therefore draw different criteria from different calibrated fits. When repeated-chain results are available, the table uses the pre-specified repeated-chain aggregate identified in the metric-level record. Q–DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger marks a warning in the supporting diagnostic record; a double dagger marks a failed diagnostic check.

Model	$p = 0.05$			$p = 0.25$			$p = 0.50$		
	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss	Fit RMSE	Forecast MAE	Forecast check loss
DQLM	2.623	3.003	**1.505**	4.740	2.487	4.540	2.226	1.932	5.556
exDQLM	**2.554**	23.91	5.857	1.381	8.941	5.649	2.408	2.207	5.607
Q–DESN AL–RHS†	3.154	3.006	1.564	1.855	**0.9122**	**4.467**	1.271	**0.9074**	**5.441**
Q–DESN exAL–RHS‡	3.354	**2.913**	1.527	**1.380**	1.819	4.488	**0.9177**	2.562	5.610

Table 6: MCMC single-quantile fit-and-forecast comparison for the Gaussian mixture simulation family. The Q–DESN entries form a fixed case-specific metric-level summary and can therefore draw different criteria from different calibrated fits. When repeated-chain results are available, the table uses the pre-specified repeated-chain aggregate identified in the metric-level record. Q–DESN preprocessing is estimated only from the training window. Forecast criteria average rolling-origin lead-target pairs over a held-out window of length 1000 using leads 1–30 and origin stride 30. Lower values are better, and boldface marks the best displayed value within each target level and criterion. A dagger marks a warning in the supporting diagnostic record; a double dagger marks a failed diagnostic check.

Figure 2 provides a cross-family view of the same comparisons. Its shading is normalized only within a fixed family, quantile level, and metric; the printed cell values retain the absolute scale. The colors therefore show relative departures from the best displayed value without combining RMSE, MAE, and check loss into a single criterion.

Displayed validation comparison. The result tables are derived from the validation records without altering the underlying source tables. The reproducibility record stores the source identities, table-generation hashes, row counts, compact replacements, and remaining diagnostic qualifications. The displayed MCMC tables should therefore be interpreted as the reported validation study for this fixed dynamic design, not as a claim about larger-window MCMC runs, alternative fit sizes, multiple dynamic roots, or an unreported reservoir hyperparameter search. Variational Bayes panels from the same validation record are reported in the supplement as computational companion evidence.

A coherent five-chain sensitivity for the Gaussian family at $p = 0.25$ is reported in the supplement. It assesses whether the selected Q–DESN behavior persists across independently seeded chains, while remaining separate from the primary metric-specific summary because the two summaries use different estimators.

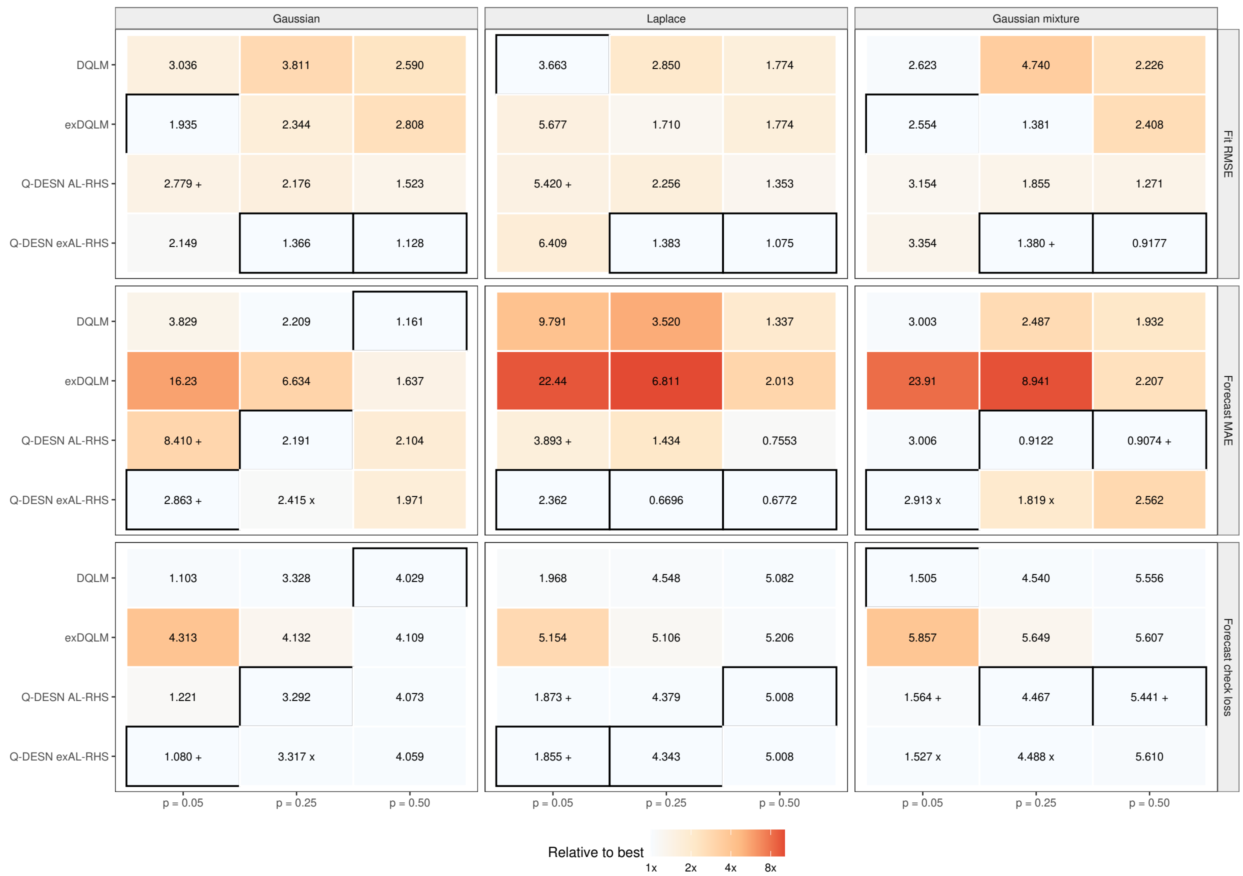

Figure 2: MCMC performance across the independent single-quantile simulation families. Columns identify the data-generating family and rows identify the criterion. Each cell prints the absolute criterion value, while shading gives its ratio to the lowest value among the four models for that family, quantile level, and criterion. Outlined cells are the within-comparison minima; darker cells indicate larger multiplicative departures. Plus signs and crosses mark values whose supporting diagnostic record contains a warning or failed diagnostic check, respectively. Because the displayed record is a metric-specific summary, cells need not originate from a common specification or replicate seed. All criteria are lower-is-better. Forecast criteria use the held-out window of length 1000 with leads 1–30 and origin stride 30.

7.6 Joint Multi-Quantile Validation with a Pre-Specified Feature Design

Pre-specified validation design. The second simulation study assesses whether a shared quantile-vector regression can recover several conditional quantile paths simultaneously under a feature design fixed before evaluation. Candidate reservoir and prior specifications are selected separately for each mechanism and model using VB or VB-LD. A replicated robustness experiment then refits the 32 scenario–model specifications on 50 independently generated data-generating realizations per comparison, for 1,600 variational fits in total. The reported evidence is a balanced MCMC validation analysis of the same four regression–likelihood combinations: Joint QDESN RHS, Independent QDESN RHS, Joint exQDESN RHS, and Independent exQDESN RHS. Each MCMC run is initialized from its corresponding variational fit and retains the scenario-specific specification fixed before sampling. All comparisons concern posterior quantile-grid paths; they do not validate a scalar predictive density formed by multiplying or mixing the component working likelihoods. The updated validation table preserves all 16 AL rows. For the 16 exAL rows, it uses the prospectively selected exact-M0 evidence in 11 scenario–model comparisons whose reported quantile functionals satisfy the stability criterion and retains five previously verified rows where that functional is unstable. The 11 M0 rows retain their mixing qualification. Across all 16 prospective M0 comparisons, M0 improves fit-window MAE relative to the historical exAL result, but improves forecast MAE in only five, three of which are functionally stable enough to be selected for reporting.

The selected regression is fit across

$$\tau \in \{0.05, 0.10, 0.25, 0.50, 0.75, 0.90, 0.95\}.$$

The balanced MCMC comparison grid uses eight synthetic data-generating mechanisms, a pre-specified feature design, and a 500-observation fit window. The three bridge mechanisms use Gaussian, Laplace, and Gaussian-mixture innovations. The five stress mechanisms introduce asymmetric tails, persistent heavy tails, Student-t errors, regime shifts, and nonlinear reservoir-friendly dynamics. The resulting grid contains 32 scenario–model MCMC rows. Fitted and forecast quantiles are compared with the known conditional quantile paths; realized observations are scored by check loss and finite-grid aCRPS; and hit-rate errors summarize marginal calibration. Reported scores use the pre-specified monotone rearrangement, while raw adjacent-level crossings before that rearrangement are retained as numerical coherence diagnostics.

Scenario-level results. Table 7 summarizes the balanced MCMC validation analysis at the scenario level. Entries report forecast MAE against the oracle quantile grid, with fit-window MAE and raw forecast-window crossing counts in parentheses; boldface marks the lowest within-scenario forecast MAE. Joint QDESN RHS is lowest in the Laplace, nonlinear, and normal mechanisms. Independent QDESN RHS is lowest in the asymmetric-tail, regime-shift, and Student-t location-scale mechanisms, while Independent exQDESN RHS is lowest in the Gaussian-mixture and persistent-heavy-tail mechanisms. Joint exQDESN RHS is not the forecast-MAE minimum in this validation comparison. These numerical lowest-scoring methods are descriptive because complete best-versus-runner-up

Scenario	Mechanism	Joint QDESN AL–RHS	Independent QDESN AL–RHS	Joint exQDESN exAL–RHS	Independent exQDESN exAL–RHS
Asymmetric Laplace Tail	Asymmetric Laplace; seasonal AR(1) location-scale	0.099 (0.084; 1)	**0.083 (0.088; 7)**	0.093 (0.081; 0)	0.089 (0.077; 0)
Gaussian-Mixture Bridge	Gaussian mixture; seasonal AR(1) location-scale	0.096 (0.097; 0)	0.099 (0.101; 0)	0.106 (0.096; 0)	**0.094 (0.095; 0)**
Laplace Bridge	Laplace; seasonal AR(1) location-scale	**0.085 (0.089; 0)**	0.094 (0.091; 7)	0.111 (0.113; 0)	0.106 (0.117; 0)
Nonlinear Reservoir Friendly	Gaussian mixture; nonlinear reservoir dynamics	**0.104 (0.109; 0)**	0.116 (0.112; 1)	0.157 (0.151; 0)	0.149 (0.143; 0)
Normal Bridge	Gaussian; seasonal AR(1) location-scale	**0.081 (0.082; 0)**	0.082 (0.082; 2)	0.115 (0.097; 0)	0.104 (0.101; 0)
Persistent Heavy Tail	Student-t; persistent seasonal AR(1) dynamics	0.131 (0.120; 0)	0.124 (0.118; 0)	0.130 (0.103; 0)	**0.106 (0.100; 0)**
Regime Shift	Student-t; regime-shift location-scale	0.153 (0.085; 0)	**0.096 (0.091; 6)**	0.185 (0.090; 0)	0.113 (0.087; 0)
Student-t Location-Scale	Student-t; seasonal AR(1) location-scale	0.080 (0.062; 0)	**0.071 (0.065; 1)**	0.116 (0.093; 0)	0.099 (0.092; 0)

Table 7: Scenario-level MCMC validation analysis for the joint multi-quantile validation study. Entries are forecast MAE with fit-window MAE and the number of raw forecast-window adjacent-level crossings in parentheses, written as forecast MAE (fit MAE; raw crossings). Boldface marks the lowest forecast MAE within each pre-specified eight-mechanism scenario. QDESN uses the AL working likelihood, exQDESN uses the exAL working likelihood, and RHS denotes the regularized horseshoe prior. All scores use the monotone quantile-grid reporting rule; raw crossings are retained only as pre-rearrangement diagnostics. Updated exAL comparisons are reported when finite-path and quantile-stability criteria are satisfied; otherwise previously verified MCMC values are used.

Monte Carlo error is unavailable whenever a retained historical row is involved.

Coherence diagnostics and interpretation. Fit recovery gives a complementary result: Joint QDESN RHS has the lowest MAE in five of the eight fit windows, with Independent exQDESN RHS lowest in the three remaining cases. Quantile coherence also distinguishes the regression structures. Independent QDESN RHS accounts for 24 of the 25 raw forecast-window crossings; Joint QDESN RHS accounts for one, and both exAL rows account for none. All 32 rows have zero crossings after the monotone rearrangement. The replicated VB experiment further gives lower median paired forecast MAE to AL than exAL for both regression structures in every mechanism, although the frequency and size of the differences vary by scenario. Together, these results retain Joint QDESN RHS as the principal joint-regression reference in this design. They do not support a claim that the additional exAL shape parameter, or the M0 update, uniformly improves forecast quantile-path accuracy. The supplement reports the computational protocol, diagnostic criteria, metric-specific lowest-scoring methods, and replicated AL–exAL contrasts.

8 Application: GloFAS Retrospective Streamflow Quantile Forecasting

Retrospective case study. The empirical application considers a medium-range retrospective streamflow quantile-forecasting case using the Global Flood Awareness System (GloFAS) and a reference gauge record. GloFAS is a global hydrological forecast and monitoring system jointly developed by the European Commission and the European Centre for Medium-Range Weather Forecasts (ECMWF) and operated within the Copernicus Emergency Management Service (Alfieri et al., 2013; Copernicus Emergency Management Service, 2026). Its operational forecast and reforecast products provide ensemble river-discharge information for flood early warning and forecast verification (Harrigan et al., 2023). Previous calibration work has focused on hydrological model parameters using daily streamflow observations (Hirpa et al., 2018); here Q–DESN is used as a statistical quantile-regression correction for one forecast system and one audited forecast origin. The case is intentionally not described as an operational calibration study: the selected specification is retrospective, uses a blended future-covariate information set, and is evaluated at a single origin.

8.1 Data and Forecast-Origin Protocol

Let T denote a forecast origin and let H be the issued forecast horizon. The historical reference record is y_t, the transformed USGS streamflow through time T. The retrospective GloFAS product over the same period is g_t^{ret}, and $g_{T+h,m}^{\text{ens}}$ denotes GloFAS ensemble member m issued at origin T for horizon $h = 1, \ldots, H$. The values $y_{T+1}, \ldots, y_{T+H}$ are not available at the forecast origin. They are held out during prediction and used only later for validation.

Retrospective information set. The reported analysis uses the forecast origin 2022-12-25 and scores seven fitted quantile levels over 28 issued target dates. The application is therefore an audited single-origin retrospective case rather than a basin-wide, multi-origin, or deployable operational evaluation. Its response values are held out for scoring, but the selected covariate policy includes realized-history and blended Global Ensemble Forecast System (GEFS) forecast information with realized-future weather contributions for precipitation and soil-moisture covariates. This restriction is intentional: it keeps the data extraction, forecast-origin alignment, quantile-reporting protocol, and selection record reproducible before broader comparisons are reported.

8.2 Application Model

Persistence-anchored discrepancy. For $t \leq T$, the application uses two deterministic Q–DESN feature maps. The reference block $\boldsymbol{x}_t^{\text{ref}}$ is driven by the transformed reference history and the selected covariate policy. The discrepancy block $\boldsymbol{x}_t^{\text{disc}}$ is driven by the retrospective discrepancy $g_t^{\text{ret}} - y_t$, using the same audited preprocessing, lag, reservoir, reducer, and washout conventions. For $t > T$, the selected retrospective specification uses the persistence-anchored innovation rule: the forecast-window discrepancy feature is anchored at the last historical retrospective discrepancy and the discrepancy regression models an innovation around that persistence baseline. The empirical issued-ensemble quantile is retained as a raw GloFAS comparator and as part of the forecast-system evidence table; it is not the selected forecast-window discrepancy-reservoir input.

For a fixed quantile level p_0, the reference-process quantile is represented by one Q–DESN regression and the GloFAS discrepancy by a second regression,

$$q_{p_0,t}^{\text{ref}} = \boldsymbol{x}_t^{\text{ref}\top}\boldsymbol{\beta}_{p_0}, \qquad d_{p_0,t}^{\text{glo}} = \boldsymbol{x}_t^{\text{disc}\top}\boldsymbol{\alpha}_{p_0}, \qquad q_{p_0,t}^{\text{glo}} = q_{p_0,t}^{\text{ref}} + d_{p_0,t}^{\text{glo}}.$$

The path $q_{p_0,t}^{\text{ref}}$ is the target reference-process quantile, while $d_{p_0,t}^{\text{glo}}$ is the quantile-specific discrepancy between GloFAS and the reference process. The retrospective and ensemble GloFAS products inform the forecast-system quantile path $q^{\text{glo}}_{p_0,t}$; the reported target remains $q^{\text{ref}}_{p_0,T+h}$.

At forecast origin T, the issued GloFAS ensemble supplies forecast-system alternatives at each covered target date, not a direct forecast of the reference-process quantile. The selected application output is a VB quantile-regression summary rather than response-level posterior-predictive sampling. For variational draw or summary index $s = 1, \ldots, S$, the construction records the reference quantile, the persistence-anchored discrepancy innovation, and the resulting forecast-system quantile:

$$q^{\text{ref},(s)}_{p_0,T+h} = \boldsymbol{x}^{\text{ref},(s)\top}_{T+h} \boldsymbol{\beta}^{(s)}_{p_0}, \qquad q^{\text{glo},(s)}_{p_0,T+h} = q^{\text{ref},(s)}_{p_0,T+h} + d^{\text{ret}}_T + \boldsymbol{x}^{\text{disc},(s)\top}_{T+h} \boldsymbol{\alpha}^{(s)}_{p_0}.$$

Here d^{ret}_T is the last historical retrospective discrepancy on the transformed scale. Posterior means, medians, intervals, and monotone multi-quantile summaries are computed after this quantile construction. A recursive alternative based on $\hat{q}^{\text{ens}}_{p_0,T+h} - y_{T+h}$ exists in the code base, but it is not the selected retrospective specification reported in this article.

The working-likelihood specification is

$$\begin{aligned}
y_t \mid q^{\text{ref}}_{p_0,t}, \sigma_{\text{ref}}, \gamma_{\text{ref}} &\sim \text{exAL}_{p_0}(q^{\text{ref}}_{p_0,t}, \sigma_{\text{ref}}, \gamma_{\text{ref}}), \\
g^{\text{ret}}_t \mid q^{\text{glo}}_{p_0,t}, \sigma_{\text{glo}}, \gamma_{\text{glo}} &\sim \text{exAL}_{p_0}(q^{\text{glo}}_{p_0,t}, \sigma_{\text{glo}}, \gamma_{\text{glo}}), \\
g^{\text{ens}}_{T+h,m} \mid q^{\text{glo}}_{p_0,T+h}, \sigma_{\text{glo}}, \gamma_{\text{glo}} &\sim \text{exAL}_{p_0}(q^{\text{glo}}_{p_0,T+h}, \sigma_{\text{glo}}, \gamma_{\text{glo}}), \quad h = 1, \ldots, H, \quad m = 1, \ldots, M.
\end{aligned}$$

This display gives the general exAL form. The AL specialization used by the selected AL analysis omits s-latent variables and γ blocks, and replaces the exAL constants by the corresponding AL constants at the fitted level. The issued ensemble members enter through a working likelihood that treats them as conditionally independent and exchangeable given the forecast-system quantile path and source-specific likelihood parameters. ECMWF ensemble forecasts use perturbed initial states and model perturbations to produce distinct forecast alternatives (European Centre for Medium-Range Weather Forecasts, 2026); the conditional-independence factorization is therefore a modeling approximation for the Q–DESN regression layer, not a validated dependence model for the physical ensemble. In exAL variants, the default application model shares σ_{glo} and γ_{glo} between retrospective and issued-ensemble GloFAS rows; under the AL specialization, only the corresponding σ_{glo} block is shared.

For this retrospective analysis, the seven fitted quantile levels are 0.05, 0.15, 0.35, 0.50, 0.65, 0.80, and 0.95. They are fitted independently with the AL special case and the VB approximation. This choice is deliberate: the streamflow application is intended as a fast retrospective quantile-regression analysis, so the reported run uses Q–DESN under AL rather than exQDESN under exAL. This application therefore uses the independent-fit and monotone-rearrangement analysis, not the joint quantile-vector regression of Section 3.3. The variational quantile summaries are then passed through the monotone rearrangement. The selected reported output uses the following reporting convention: No reporting-time spread calibration is applied; the displayed bands are synthesized from the fitted quantile models and then monotonized. This convention does not alter the reservoir features, likelihood, priors, or fitted regression coefficients.

The selected fit uses separate DESN feature maps for the reference quantile path and the GloFAS discrepancy path. Both maps have depth $D = 1$, layer widths (300), reducer widths (*none*), memory 360, washout 500, spectral radii (0.95), recurrent sparsity (0.03), and input inclusion probabilities (1). The reference map uses leak rates (0.1) and global/bias input scales 0.18/0.18, and reservoir seed 20260512; the discrepancy map uses leak rates (0.1), global/bias input scales 0.18/0.18, and reservoir seed 20261521. Forecast-window discrepancy states use the persistence-anchored innovation transition.

The two coefficient blocks use separate regularized-horseshoe global scales, $\tau_{0,\mathrm{ref}} = 0.1$ and $\tau_{0,\mathrm{disc}} = 0.001$, respectively. A configuration snapshot and selection record document the data version, preprocessing choices, selected files, and fitted specification for this retrospective analysis.

8.3 Retrospective Application Results

The selected fit was audited over 12,495 observed dates through the forecast origin. After monotone rearrangement, its observational-window aCRPS was 0.0445, and empirical 90% interval coverage was 0.906. This evidence governs the specification choice; the forecast window remains a held-out assessment. The retrospective audit also identified occasional extreme fitted values in the $p_0 = 0.95$ path, so the upper-tail component remains subject to targeted sensitivity analysis.

Forecast-window scores. Table 8 reports the forecast scores for the reported retrospective analysis. Relative to raw GloFAS, Q–DESN reduces the mean check loss from 0.7639 to 0.5654, the interval score from 25.2028 to 10.8032, and aCRPS from 1.4424 to 1.1319. Mean empirical interval coverage increases from 0.000 to 0.357, but the Q–DESN value remains far below the nominal 90% level. These summaries describe one audited forecast origin under a blended retrospective information set and should be read as a single-origin score improvement, not as multi-origin operational validation or evidence of calibrated predictive intervals.

Table 8: Forecast scoring for the GloFAS retrospective analysis. Scores are computed on the transformed streamflow scale for the forecast origin 2022-12-25. Lower check loss, interval score, and aCRPS are better; aCRPS is the finite-grid trapezoidal integrated check-loss score over the seven fitted levels. Mean coverage reports empirical coverage for the nominal 90% reported interval and is not a calibration claim. The final column gives the relative reduction in mean check loss.

Model	Horizons	Check	Interval	aCRPS	Coverage	Check reduction
Q–DESN correction	28	0.5654	10.8032	1.1319	0.357	26.0%
Raw GloFAS	28	0.7639	25.2028	1.4424	0.000	Reference

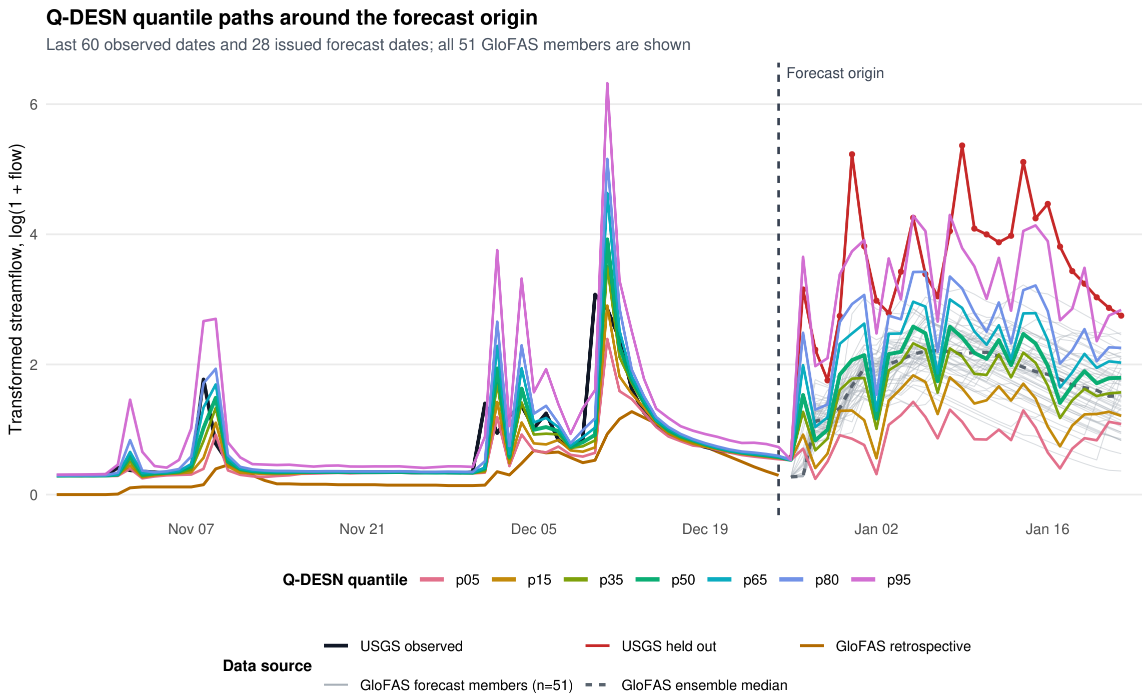

Figure 3: Discrepancy-corrected Q–DESN quantile paths for the GloFAS retrospective analysis. The figure shows the final 60 observed dates through the forecast origin (vertical dashed line) and the subsequent 28-day issued forecast window. Black denotes observed USGS streamflow before the origin, red denotes the held-out USGS measurements, and orange denotes retrospective GloFAS. Thin gray paths are all 51 issued GloFAS ensemble members; their median is the dashed gray path.

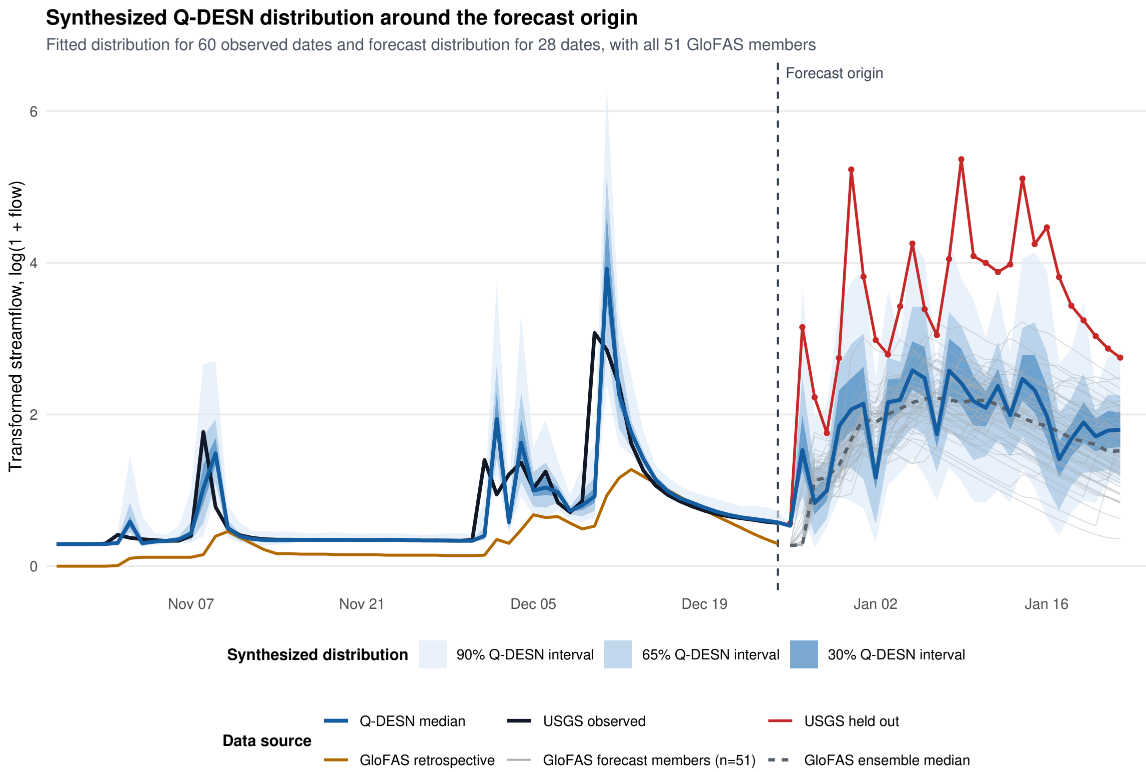

Figure 4: Synthesized Q–DESN quantile bands for the GloFAS retrospective analysis over the final 60 observed dates and the subsequent 28-day forecast window. The nested blue bands are the 90%, 65%, and 30% quantile intervals, and the blue path is the median. The vertical dashed line marks the forecast origin. Black and red paths denote observed and held-out USGS streamflow, respectively; orange denotes retrospective GloFAS, and thin gray paths show all 51 issued GloFAS ensemble members, with their median dashed. Forecast scores use the post-origin targets in Table 8.

Reproducibility checks confirm the data extraction, selected files, score table, and VB convergence record used for this retrospective analysis. The VB fits required a median of 103 and a maximum of 356 iterations. Future model specifications can replace this fit through the same documented selection record without changing the application model statement.

9 Application: PriceFM Retrospective Electricity-Price Comparison

Benchmark target. As a second empirical case, we consider a retrospective comparison of probabilistic day-ahead electricity-price forecasts using version 4 of the PriceFM benchmark (Yu et al., 2026). The current benchmark release spans 1 January 2022 through 1 January 2026. The reported target is original-unit quantile scoring on the PriceFM paper quantile grid under fold-aligned test timestamps. The released data cover European bidding regions at 15-minute resolution and include day-ahead price, load, solar, and wind forecasts. PriceFM itself is a graph-informed foundation model: it uses region-level exogenous information and a sparse graph mask derived from the European transmission topology. Our reported comparison is a 114 region–fold comparison against fold-aligned PriceFM predictions, not a deployable operational forecast exercise or a claim about the full paper-wide PriceFM evaluation.

9.1 Comparison Protocol

The reported benchmark contains 114 region/fold evaluations, spanning 38 European bidding regions and 3 rolling folds. Each evaluation is scored on the same fold-aligned test timestamps used by the PriceFM predictions. The scoring metric is original-unit average quantile loss (AQL) on the PriceFM paper quantile grid $0.10, 0.25, 0.45, 0.50, 0.55, 0.75, 0.90$. Throughout this section, Δ AQL denotes Q–DESN AQL minus PriceFM AQL. Negative values indicate lower Q–DESN AQL; positive values indicate lower PriceFM AQL.

Retrospective covariates. The benchmark panel is assembled from audited analysis records and previously computed paper-quantile score files. This assembly step is assembly-only: it does not fit new models and it does not alter the PriceFM predictions. Across the benchmark, 58 selected region/fold evaluations use own-region predictors and 56 selected region/fold evaluations use graph-derived PriceFM neighborhood summaries. Both input policies use retrospectively observed target-region load, solar, and wind lead covariates in the comparison; graph-derived rows add retrospectively observed neighboring-region lead summaries. The score comparison is therefore retrospective and fold-aligned, not an origin-available operational comparison. The benchmark is interpreted as an application of the selected Q–DESN quantile-regression analysis recorded by the validation records. It is not used as evidence for the joint quantile-vector RHS prior unless a future PriceFM run explicitly fits and selects that joint regression under the same fold-aligned protocol.

PriceFM reports two model-comparison protocols. Its Table II is a full-shot evaluation, whereas its Table III evaluates leave-one-region-out zero-shot generalization. Because the Q–DESN specifications here are fitted and selected with observations from each target region, the full-shot table is the relevant paper-level analogue. The zero-shot table is not used to rank the present region-specific fits.

9.2 Full-Surface Results

Aggregate comparison. Across the aligned benchmark, Q–DESN has lower AQL in 66 of 114 region/fold evaluations (57.9%), 12 evaluations are near-equivalent under the pre-specified near-equivalence threshold, and PriceFM has lower AQL in 36 evaluations. The mean Q–DESN AQL is 6.826, compared with mean PriceFM AQL 7.039, giving mean Δ AQL -0.213 and median Δ AQL -0.083. The aggregate difference is therefore modest rather than uniform. The fold-level supplementary summary shows that the mean Δ AQL is most negative in fold 2, modestly negative in fold 1, and essentially neutral in fold 3. On the same 114-comparison panel, the region–fold macro mean AQCR is 2.64% for Q–DESN and 0.00% for the released PriceFM checkpoint. Mean MAE is 16.728 versus 17.274, and mean RMSE is 25.453 versus 26.336, respectively. Thus Q–DESN reduces aligned AQL, MAE, and RMSE by approximately 3.0%, 3.2%, and 3.4%, while retaining a nonzero quantile-crossing rate.

Table 9: PriceFM-aligned retrospective model comparison. The first panel reports direct region–fold macro means from the common 38-region, three-fold comparison; boldface marks the better of those two directly aligned rows. The second panel reproduces the full-shot values in Table II of PriceFM version 4 (Yu et al., 2026). The paper-reported panel uses a distinct fitted-model and aggregation pipeline and is therefore contextual rather than a head-to-head ranking against the aligned comparison panel. AQL, MAE, and RMSE are in EUR/MWh; AQCR is in percent. Rank is reproduced only for the paper-reported panel; dashes indicate that no cross-panel rank is assigned. Lower is better. The comparison uses retrospectively observed target covariate leads in all selected rows and additional retrospectively observed neighboring-region leads in graph-derived rows.

	Probabilistic		Pointwise		
Model	AQL	AQCR (%)	MAE	RMSE	Rank
Direct fold-aligned comparison (this article; 114 region/fold evaluations)					
Selected Q–DESN	**6.83**	2.64	**16.73**	**25.45**	–
Released PriceFM checkpoint	7.04	**0.00**	17.27	26.34	–
PriceFM v4 Table II (paper-reported full-shot evaluation)					
Naïve[1]	15.29	0.00	22.06	34.68	11
Naïve[2]	15.35	0.00	23.31	34.31	13
Naïve[3]	15.46	0.00	22.64	32.61	12
FEDFormer	8.22	15.33	20.15	31.75	8
PatchTST	8.06	18.21	20.20	31.59	8
iTransformer	8.24	13.96	21.03	32.11	9
TimesNet	7.98	13.42	19.48	30.94	7
TimeXer	8.30	14.77	21.94	31.88	10
GraphConv	6.61	6.88	16.81	25.97	4
GraphAttn	7.13	10.33	17.00	26.11	6
GraphSAGE	6.78	6.01	17.56	26.03	5
GraphDiffusion	6.69	5.72	16.44	25.93	2
GraphARMA	6.72	6.03	16.56	25.84	3
PriceFM	**5.80**	**0.00**	**14.28**	**22.39**	**1**

The directly aligned checkpoint is the appropriate distribution-level comparator because its predictions are aligned row by row with the Q–DESN predictions. Its AQL differs from the PriceFM paper's optimized full-shot AQL, so the two PriceFM rows are deliberately not conflated. Numerically, the aligned Q–DESN AQL lies within the range reported for the graph baselines in PriceFM Table II, but differences in fitted objects and aggregation preclude a cross-panel rank claim. The fold-level win, tie, and loss summary from the previous main-text table is retained as a supplementary diagnostic.

The input set also matters. Graph-derived covariates have the most negative average Δ AQL, whereas target-only evaluations are closer to neutral. This comparison should not be read as a causal attribution to graph features, because input sets and selected model specifications vary across region/fold evaluations. It is a diagnostic summary of where the selected analysis record is strongest on the aligned benchmark. Input-set and horizon-block breakdowns are reported in the supplement.

9.3 Horizon Diagnostics and Remaining Gaps

Horizon-block diagnostics are available for 72 of the 114 region/fold evaluations. These diagnostics preserve the same convention as the aggregate comparison: negative Δ values indicate lower Q–DESN AQL. They should be read as a partial diagnostic layer rather than as a second full-surface summary, because not every archived analysis record retained horizon-block scores.

Remaining gaps. The resulting evidence supports a conservative interpretation. In this retrospective comparison, Q–DESN is a competitive fixed-reservoir quantile-regression baseline against fold-aligned PriceFM predictions on the aligned benchmark panel, with lower mean and median AQL and lower region–fold macro MAE and RMSE in aggregate. This descriptive advantage is relative to the released checkpoint comparison, not to the separately reported optimized full-shot PriceFM result, and it is not accompanied here by a paired uncertainty interval. It is not uniformly lower than PriceFM: fold 3 is nearly tied on average, and PriceFM has lower AQL for a nontrivial subset of region/fold evaluations. Subsequent read-only sensitivity checks of those PriceFM-favored evaluations did not produce a validation-selected Q–DESN candidate whose pre-specified test evaluation improved aligned original-scale AQL, so they are treated as negative diagnostic evidence rather than as a basis for changing the article tables or figures. A new PriceFM run should therefore require a pre-specified change to the design: for example, the information set, model class, selection rule, or joint quantile-regression protocol. A fully joint PriceFM run would be needed before making application-level claims about the joint multi-quantile regression model.

10 Discussion

Q–DESN treats the fixed DESN reservoir as a nonlinear feature map and places Bayesian quantile inference on the regression coefficients. This separation is useful computationally: posterior simulation and VB update only the coefficients, likelihood, and shrinkage parameters, while the recurrent weights, reducers, scaling constants, and washout convention remain fixed. The empirical results therefore evaluate a conditional Bayesian quantile-regression analysis. They should not be interpreted as posterior averaging over reservoir architectures or as a fully Bayesian treatment of recurrent network uncertainty.

The likelihood is also deliberately limited in interpretation. The AL and exAL families are used as quantile-targeting working likelihoods. They provide posterior targets and conditionally tractable augmented updates, but credible bands from the fitted working likelihood are not automatic frequentist predictive coverage guarantees under misspecification. The simulation and application sections therefore report fit recovery, forecast scoring, empirical predictive coverage, runtime, convergence, and reproducibility diagnostics rather than relying on posterior summaries alone. Sandwich adjustments and other general-Bayes calibration checks remain important sensitivity directions, especially for composite multi-quantile likelihoods.

The multi-quantile construction makes the same distinction. Independent single-level Q–DESN fits followed by isotonic regression or rearrangement produce monotone reported quantile curves, but they do not define a joint posterior over quantile-indexed regression coefficients. The joint quantile-vector extension shares information across quantile levels through adjacent RHS slope innovations, giving a soft non-crossing mechanism. It does not impose the non-crossing inequalities as hard constraints, and its composite working likelihood is not a unique response predictive density. When exact monotonicity is required for downstream use, the reported curve should still be obtained through an ordered-intercept parameterization, projection, rearrangement, or another explicitly specified monotone reporting rule.

The validation evidence supports correspondingly limited claims. The single-quantile dynamic simulations test recovery and held-out forecasting under fixed data-generating mechanisms. The joint multi-quantile synthetic study checks coherence under known conditional quantile paths. The GloFAS application is an audited one-origin retrospective streamflow quantile-forecasting case using computationally light independent Q–DESN AL–VB fits, blended future weather covariates, monotone reporting, and an undercovered nominal 90% interval. The PriceFM comparison is a retrospective fold-aligned comparison against stored benchmark predictions with retrospectively observed lead covariates. These studies support the reported analyses under their own protocols, not a universal ranking of Q–DESN over all dynamic quantile or foundation-model alternatives.

Several extensions are natural. On the statistical side, future work should study calibrated composite-likelihood scaling, hard or probabilistic non-crossing constraints, richer covariance structures for quantile innovations, and sensitivity to reservoir scaling on bounded feature domains. On the computational side, the joint model calls for sparse precision solvers, selected covariance extraction for VB, and diagnostics comparing VB with MCMC on small joint problems. On the application side, deployable multi-origin GloFAS studies with origin-available covariates, paired PriceFM uncertainty analyses, and explicitly joint PriceFM runs are needed before the joint quantile-vector prior can be interpreted as application-level evidence rather than a synthetic validation result.

Within these boundaries, the article establishes Q–DESN as a reproducible fixed-feature Bayesian quantile-regression analysis whose inference, monotone reporting, diagnostics, and empirical validation are tied to explicit protocols.

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